



CMPT 713: Natural Language Processing

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2023-01-04

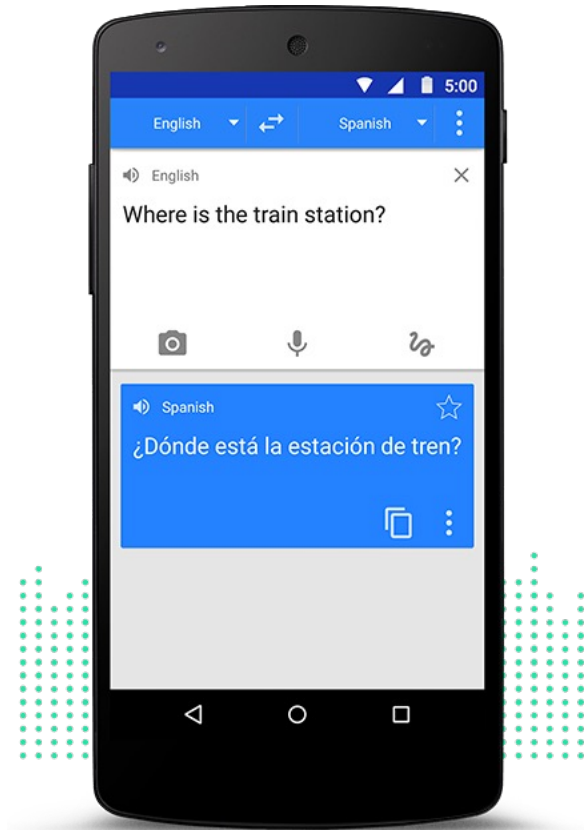


Today

- Intro to NLP
 - What is language? NLP vs computational linguistics.
 - Brief history of NLP
 - Levels of representation in language
 - Key questions of NLP
- Course Logistics



NLP is everywhere



Google translate



Virtual assistants

Information finding



when were the first pyramids built



All

Images

News

Shopping

Videos

More

Settings

Tools

About 20,300,000 results (0.67 seconds)

2630 BC

Most were built as tombs for the country's pharaohs and their consorts during the Old and Middle Kingdom periods. The earliest known Egyptian pyramids are found at Saqqara, northwest of Memphis. The earliest among these is the Pyramid of Djoser (constructed **2630 BC**–2611 BC) which was built during the third dynasty.

[The First Pyramids Built - Timeline Index](#)

www.timelineindex.com › [content](#) › [view](#)



Text to image generation

 **OpenAI** DALL-E 2

a painting of a fox sitting in a field at sunrise in the style of Claude Monet



<https://openai.com/dall-e-2/>

Try it out yourself: <https://huggingface.co/spaces/stabilityai/stable-diffusion>,
<https://www.craiyon.com/>, <https://www.midjourney.com/home/>

 **Imagen**

A cute sloth holding a small treasure chest.
A bright golden glow is coming from the chest.

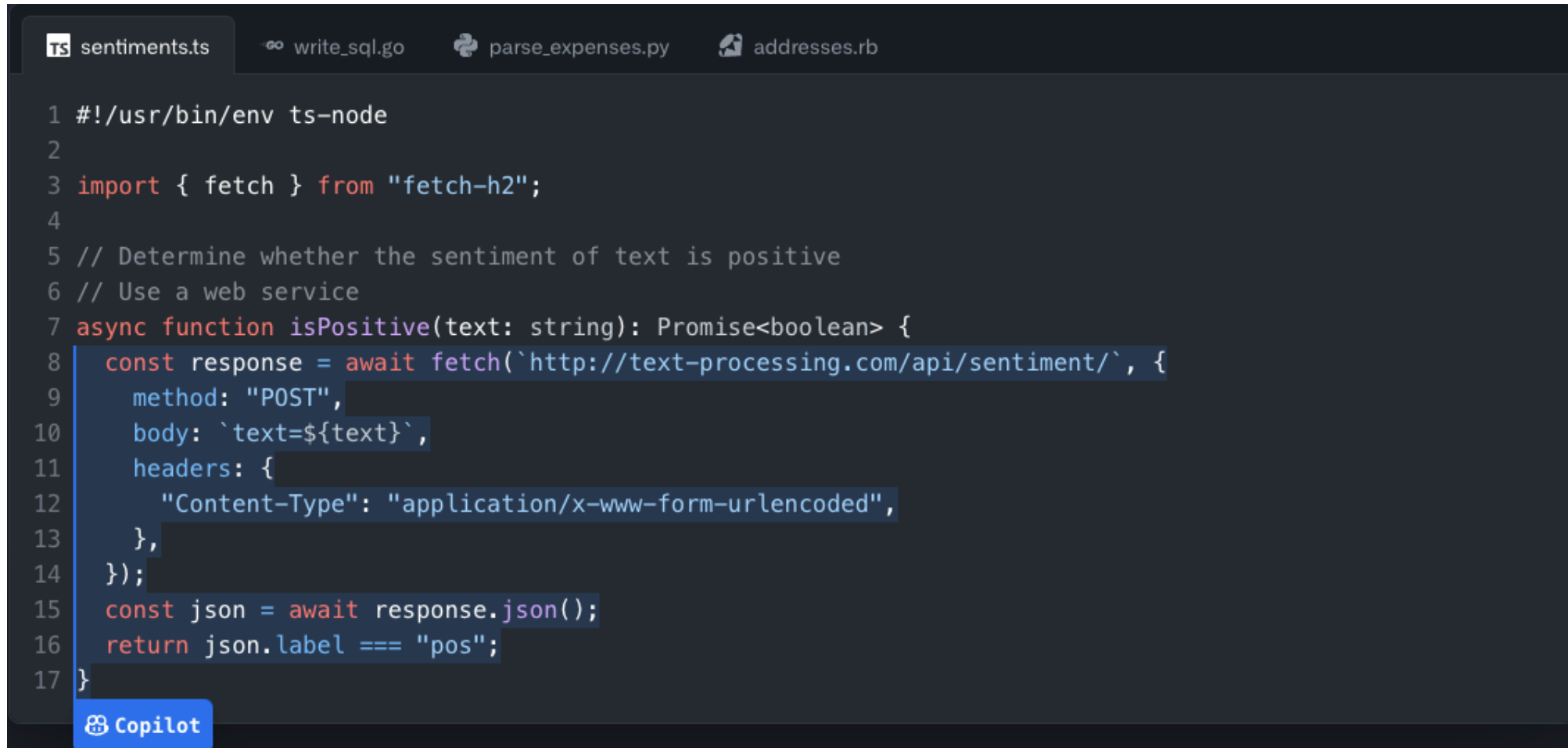


<https://imagen.research.google/>



Copilot

Generate code from text!



```
1 #!/usr/bin/env ts-node
2
3 import { fetch } from "fetch-h2";
4
5 // Determine whether the sentiment of text is positive
6 // Use a web service
7 async function isPositive(text: string): Promise<boolean> {
8   const response = await fetch(`http://text-processing.com/api/sentiment/`, {
9     method: "POST",
10    body: `text=${text}`,
11    headers: {
12      "Content-Type": "application/x-www-form-urlencoded",
13    },
14  });
15  const json = await response.json();
16  return json.label === "pos";
17 }
```

Copilot

<https://github.com/features/copilot>



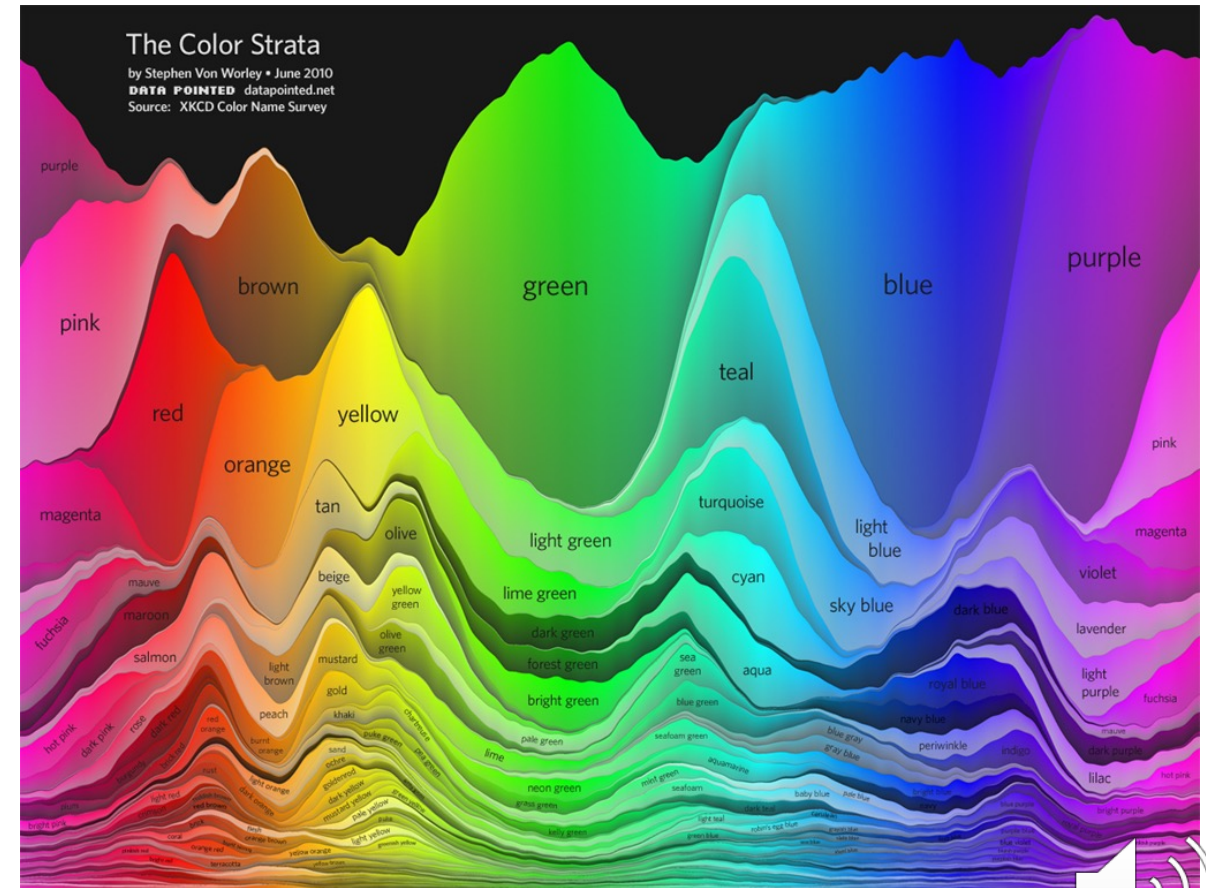
What is language?

- Language is used to communicate
 - Things, actions, abstract concepts



What is language?

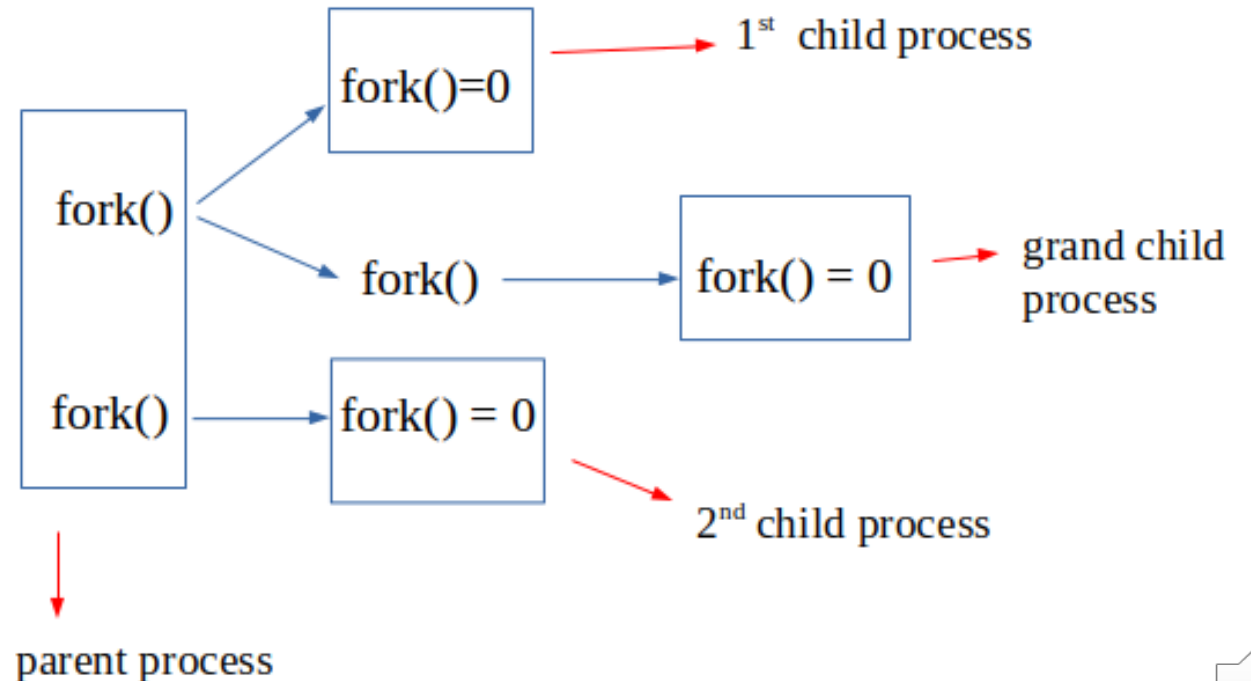
- Language puts categories on the world
 - It discretizes a continuous space



What is language?

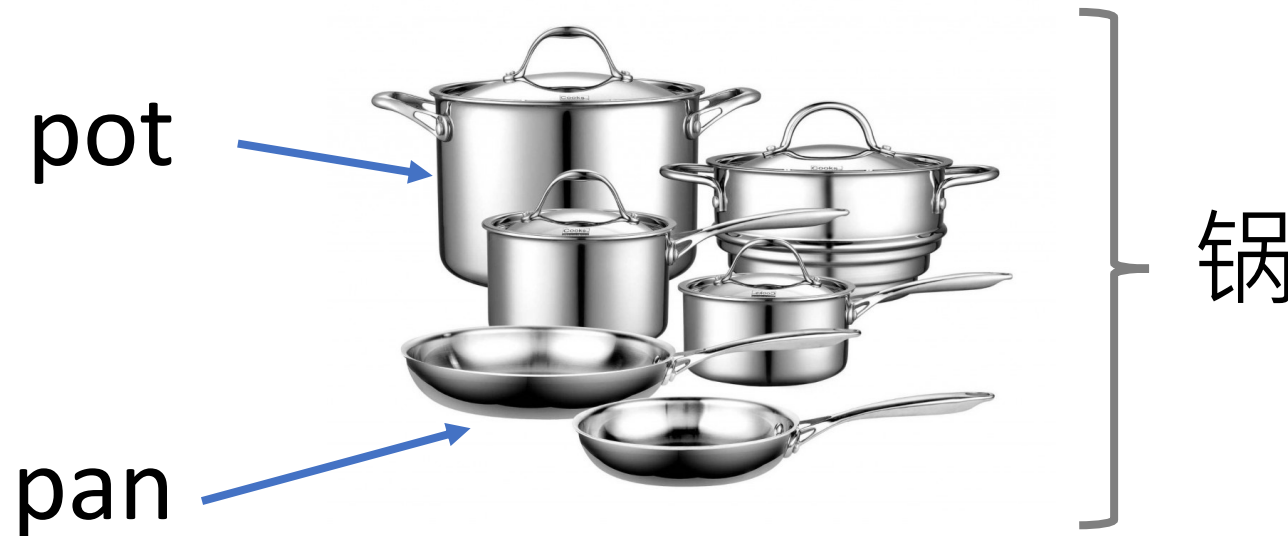
- Language picks out what is salient and important
- What concepts do we have words for?

fork



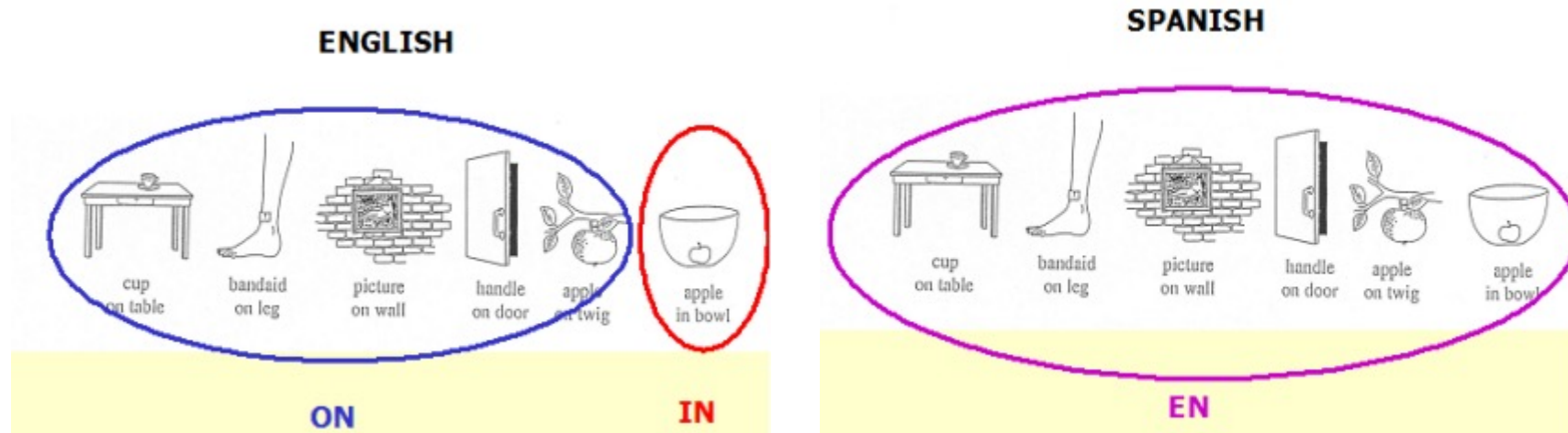
What is language?

- Language picks out what is salient and important
- What concepts do we have words for?
- Different languages have different discretization boundaries



What is language?

- Language picks out what is salient and important
- What concepts do we have words for?
- Different languages have different discretization boundaries



<http://pyersqr.org/classes/Ling731/Space2.htm>



Natural Language Processing



Building **useful system** to process language



Computational Linguistics



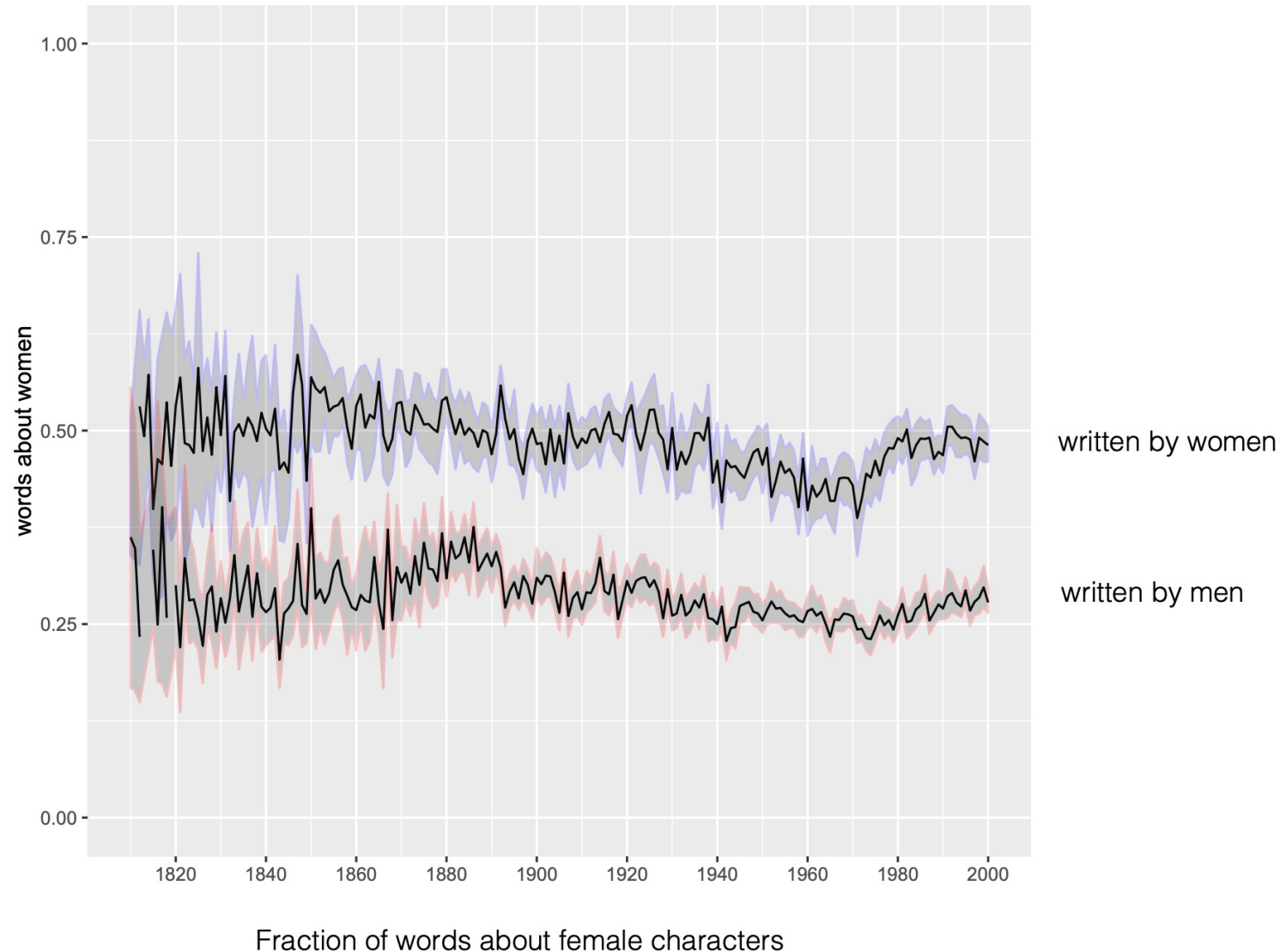
"I just had a heated conversation
with my computer."

(image credit: <https://www.enterrasolutions.com/blog/computational-linguistics-and-natural-language-processing/>)

Using computers to study **human** language



Analyzing word usage in literature

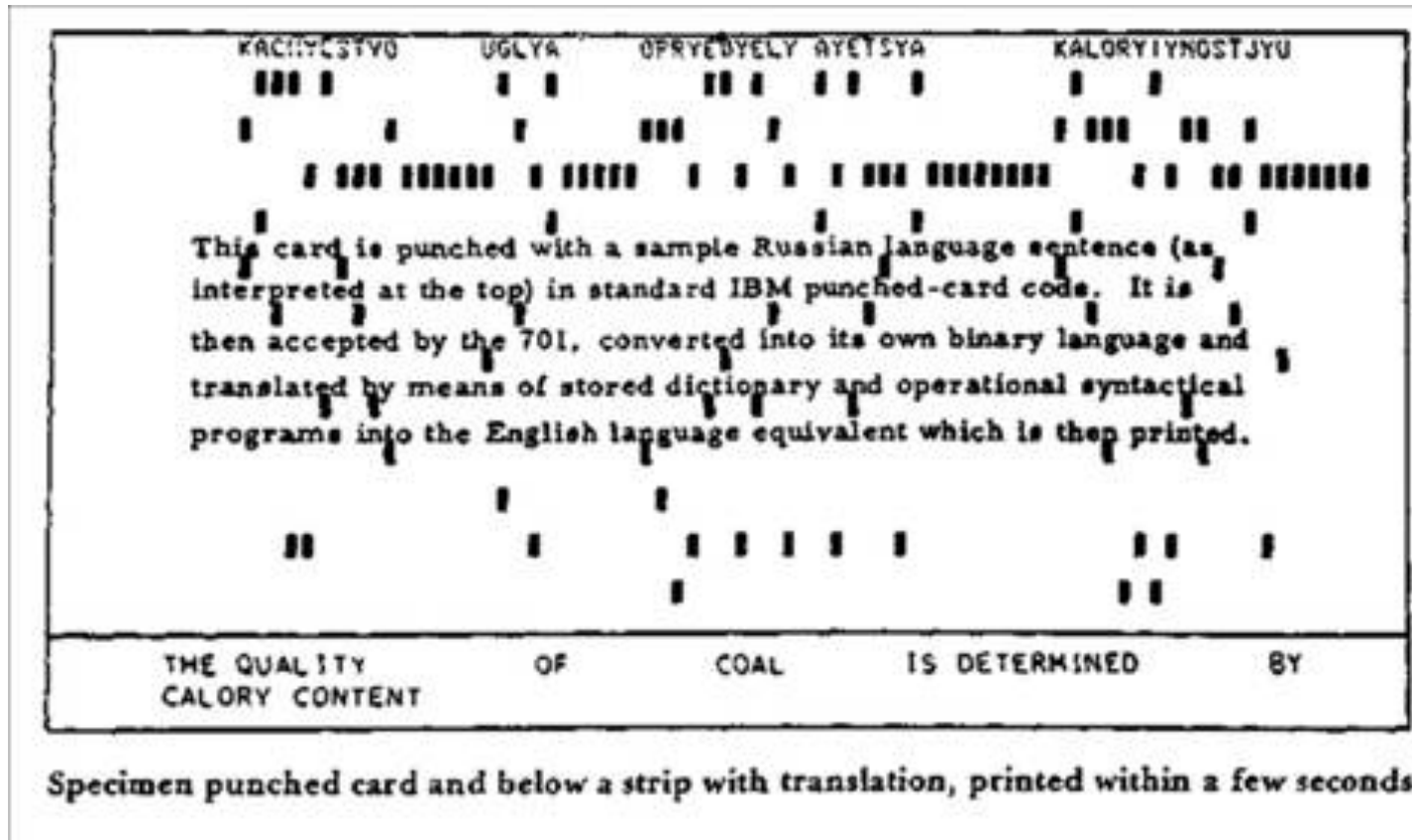


Ted Underwood, David Bamman, and Sabrina Lee (2018),
"The Transformation of Gender in English-Language Fiction," Cultural Analytics



Brief history of NLP

Beginnings



Georgetown-
IBM
experiment,
1954

“Within three or five years, machine translation will be a solved problem”



SHRDLU (Winograd, 1968)

Video of actual system: <https://www.youtube.com/watch?v=bo4RvYJYOzI>

Person: Pick up a big red block.

Computer: OK.

Person: Grasp the pyramid.

Computer: I don't understand which pyramid you mean.

Person (changing their mind): Find a block which is taller than the one you are holding and put it into the box.

Computer: By "it", I assume you mean the block which is taller than the one I am holding.

Computer: OK.

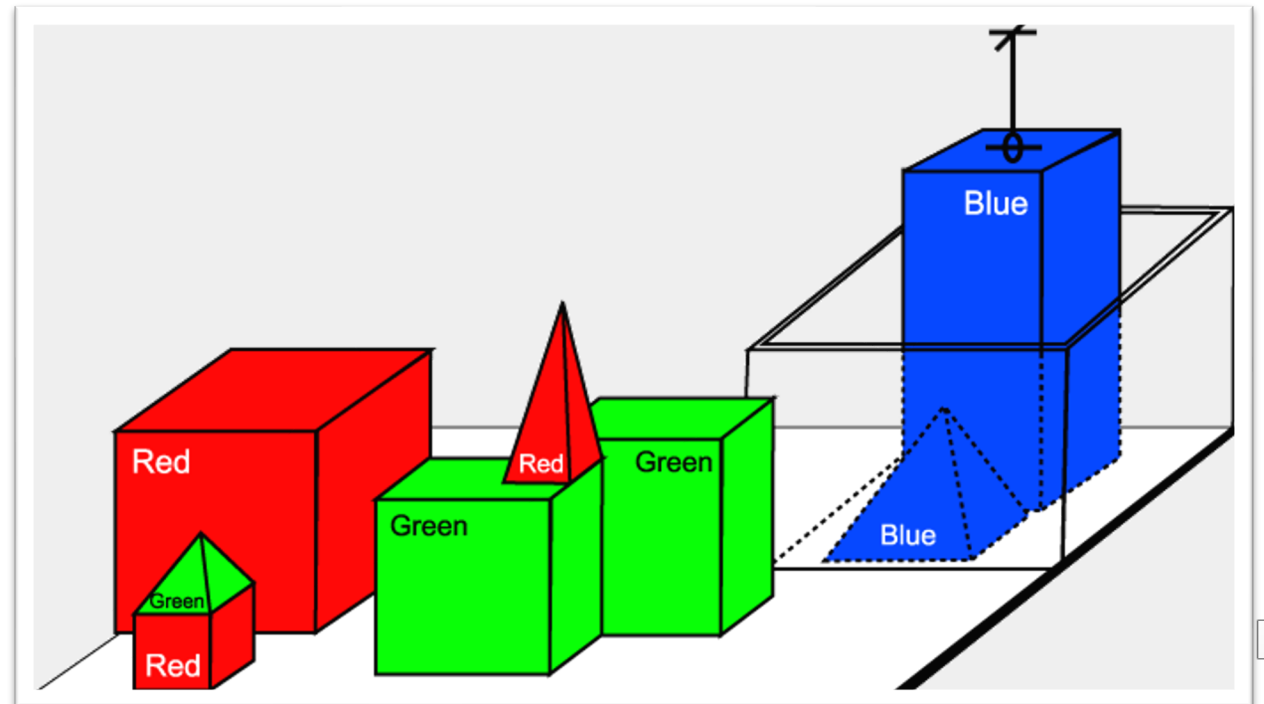
Person: What does the box contain?

Computer: The blue pyramid and the blue block.

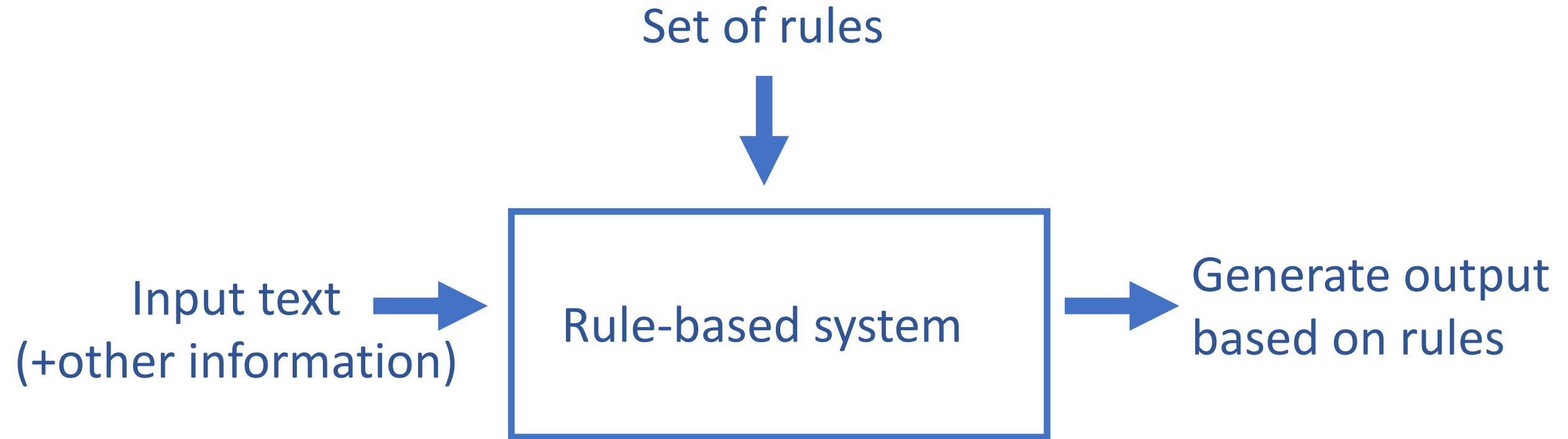
Person: What is the pyramid supported by?

Computer: The box.

Lots of rules!



Using rules for NLP



Match a pattern

- Regular expressions

*.ed => past tense verb

Contained keyword: delicious

Eliza (Weizenbaum, 1966)

Demo: <http://psych.fullerton.edu/mbirnbaum/psych101/Eliza.htm>

Men are all alike.

IN WHAT WAY

They're always bugging us about something or other.

CAN YOU THINK OF A SPECIFIC EXAMPLE

Well, my boyfriend made me come here.

YOUR BOYFRIEND MADE YOU COME HERE

He says I'm depressed much of the time.

I AM SORRY TO HEAR YOU ARE DEPRESSED

Rogarian psychologist:
reflect back what the patient said

- Set of ranked transformation **rules** based on **keywords**
- Apply rules based on matching words in sentence



Transformation rules

Keywords are ranked from specific to general

I know everybody laughed at me

- “I” is a very general keyword:

I: (I *) -> (You say you 2)

YOU SAY YOU KNOW EVERYBODY LAUGHED AT YOU

- “Everybody” is much more interesting (someone using universals like everybody/always is probably “referring to some quite specific event or person”)

WHO IN PARTICULAR ARE YOU THINKING OF?

- Implementation: keywords stored with their rank

Everybody 5 (*transformation rules*)

I 0 (*transformation rules*)

Backoff

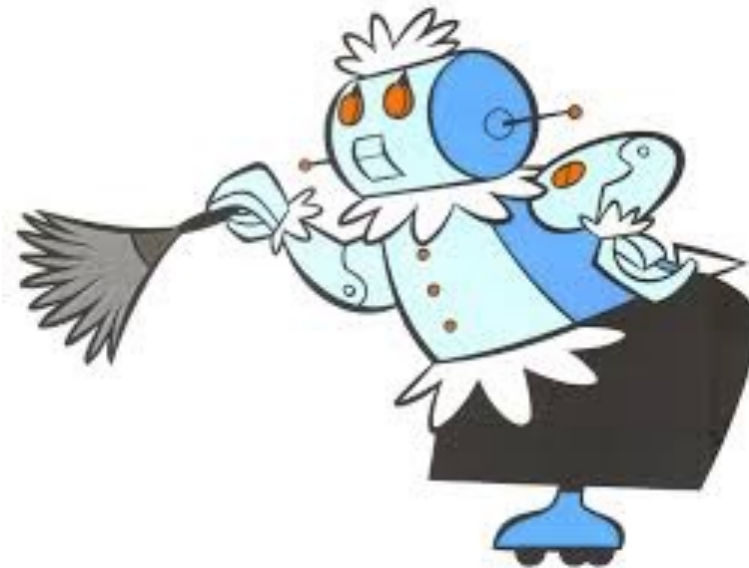
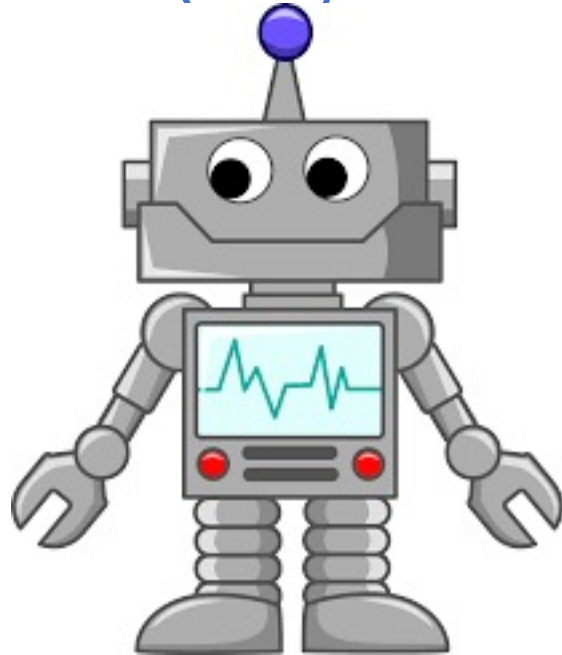
Please go on

That's very interesting

I see



Where is my block stacking or
housekeeper robot that I can talk to?
(maybe we are getting close...)



Rosie from the Jetsons





The Far Side - Gary Larson

Understanding
language is hard!



Why is natural language processing hard?



Programming Languages

C, C++, Java, Python, ...

- Unambiguous
- Fixed
- Designed
- Learnable?
- Known simple semantics

Natural Languages

French, English, Korean, Chinese, Tagalog, ...

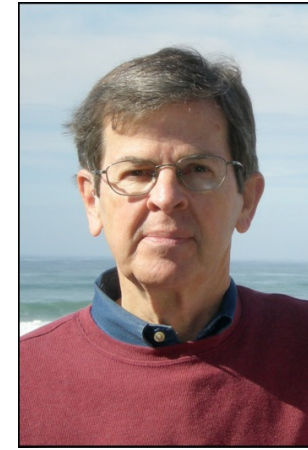
- Ambiguous
- Evolving
- Transmitted
- Learnable
- Complex semantics



Why is NLP hard?

Interpretation of language assumes a common basis of **world knowledge** and **context**

- **Ambiguous:**
 - “bank”, “bat”
 - “Milk Drinkers Turn to Powder”
- **Variation:** Many ways to say same thing
- **Context dependent:**
 - natural language is under-specified



Herb Clark

“bat’k”

Table



Counter



Context-dependence

“I put the bowl on the **table**”



“The numbers in the **table** don’t add up”

Table 8.2 Calculation of the χ^2 test on figures in table 8.1

Class (1)	Expected numbers		O - E		(O-E) ² /E	
	A (2)	B (3)	A (4)	B (5)	A (6)	B (7)
I	11.80	10.20	5.20	-5.20	2.292	2.651
II	24.67	21.33	0.33	-0.33	0.004	0.005
III	39.15	33.85	-0.15	0.15	0.001	0.001
IV	48.81	42.19	-6.81	6.81	0.950	1.009
V	30.57	26.43	1.43	-1.43	0.067	0.077
Total	30.57	134.00	0	0	3.314	3.833

$\chi^2 = 3.314 + 3.833 = 7.147$, d.f. = 4, $0.10 < P < 0.50$.

“I made her duck”

I cooked her duck (the waterfowl) for dinner

I cooked the duck (the waterfowl) belonging to her.

I created a model of a duck that she now has.

I caused her to lower her head



Some language humor (from real newspaper headlines)

Kids make nutritious snacks

Stolen painting found by tree

Miners refuse to work after death

Squad helps dog bite victim

Killer sentenced to die for second time in 10 years

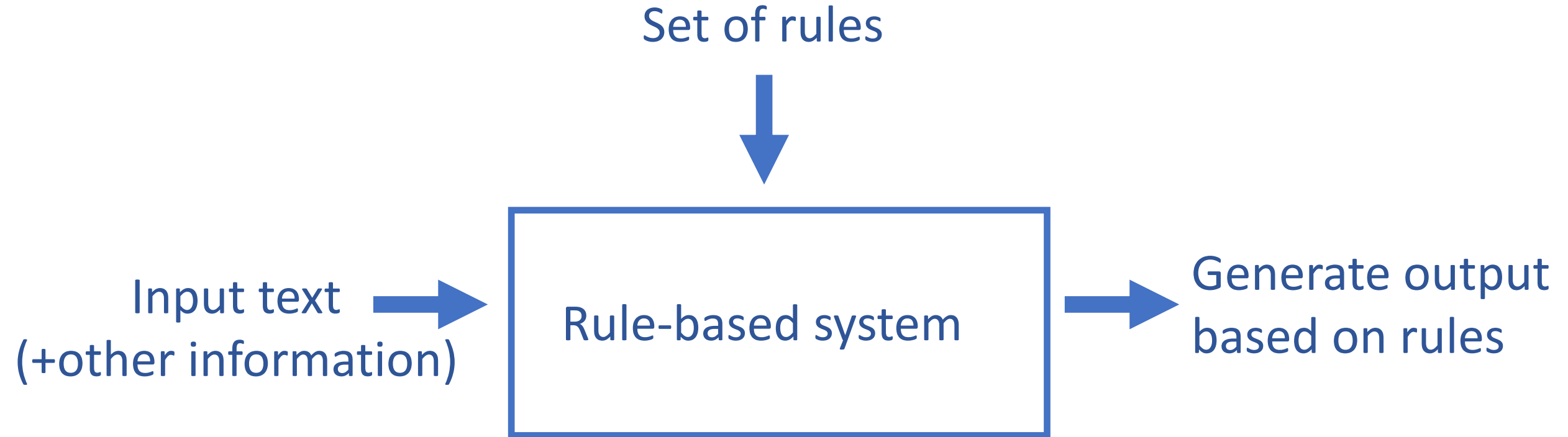
Lack of brains hinders research

Pooches and people digging newly opened park



Using rules for NLP

- Translation with rules started in the 1950s
- Eliza chatbot (1966)
- SHRDLU grounded language understanding (1968)



Rule-based Machine Translation

- Georgetown IBM translator
- Bilingual dictionary to map Russian words to their English counterparts
- But what if the word order is different? How to handle all the nuances of meaning?



1 minute video showing 1954 MT:

<https://youtu.be/K-HfpsHPmvw>

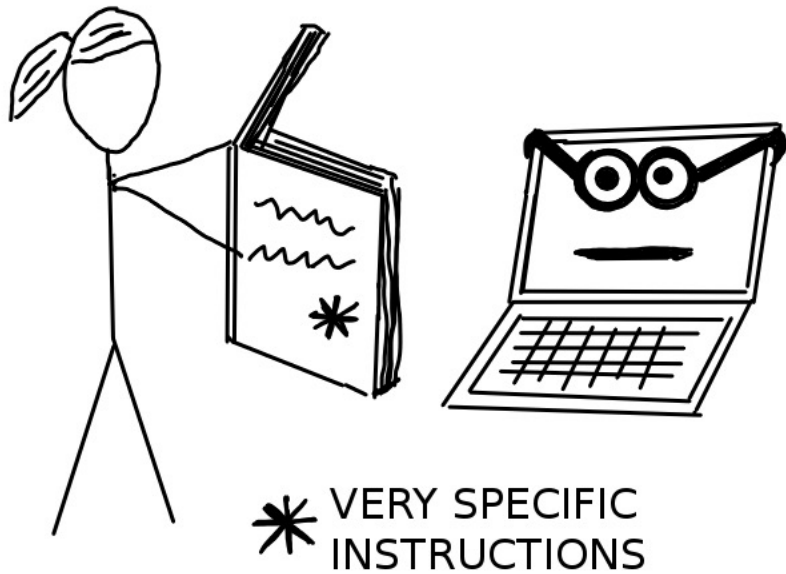
https://www.ibm.com/ibm/history/exhibits/701/701_translator.html



Coming up rules is hard!

Let's learn from data!

Without Machine Learning



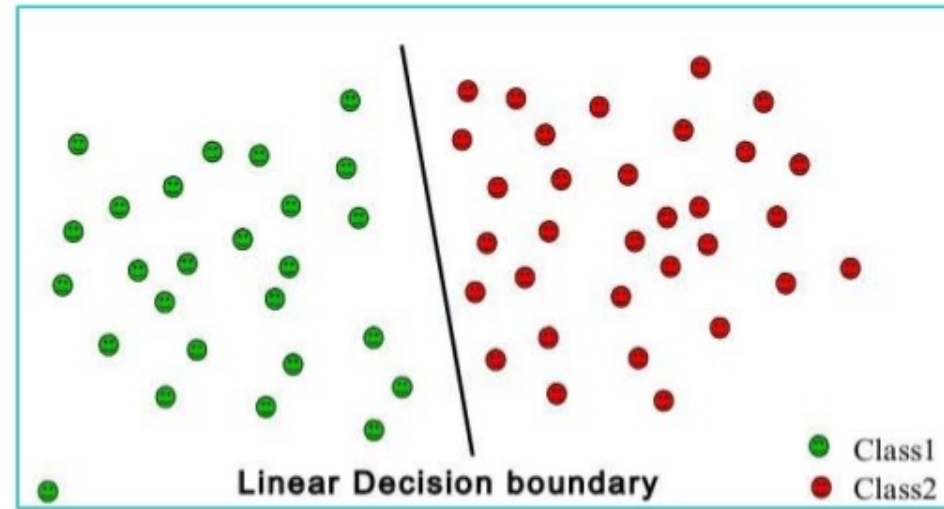
With Machine Learning



<https://christophm.github.io/interpretable-ml-book/terminology.html>

Rise of statistical learning

(late) 1980s to 2000s: Statistical MT, 2000s-2014: Statistical Phrase-Based MT

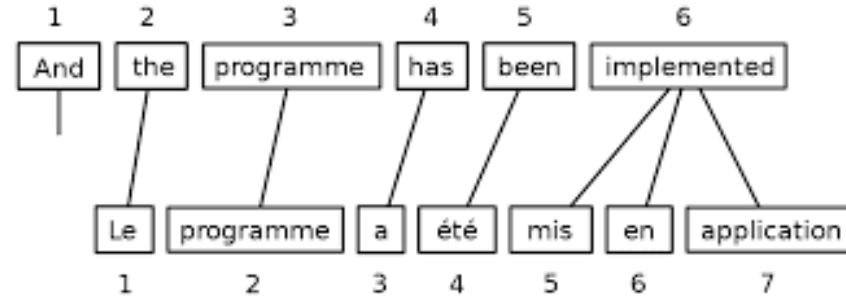


- Use of machine learning techniques in NLP
- Increase in computational capabilities
- Availability of electronic corpora
- Data-driven evaluation of models

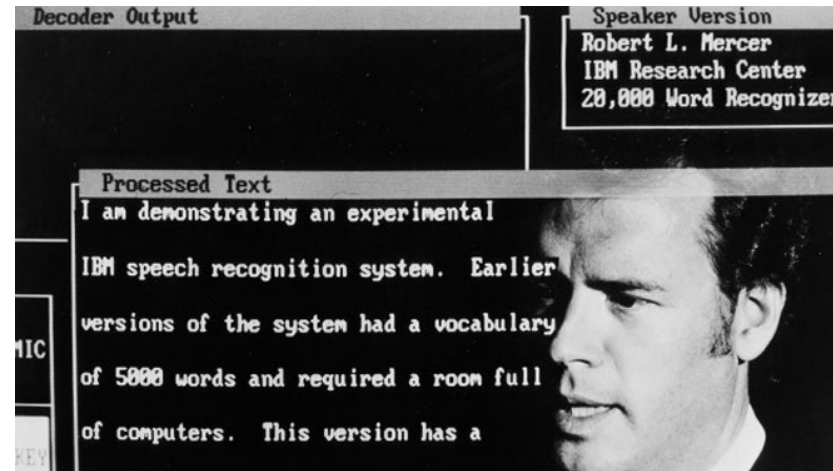


Rise of statistical learning

(late) 1980s to 2000s: Statistical MT, 2000s-2014: Statistical Phrase-Based MT



IBM Models
for translation



Speech
recognition

Anytime a linguist leaves the group the (speech) recognition rate goes up

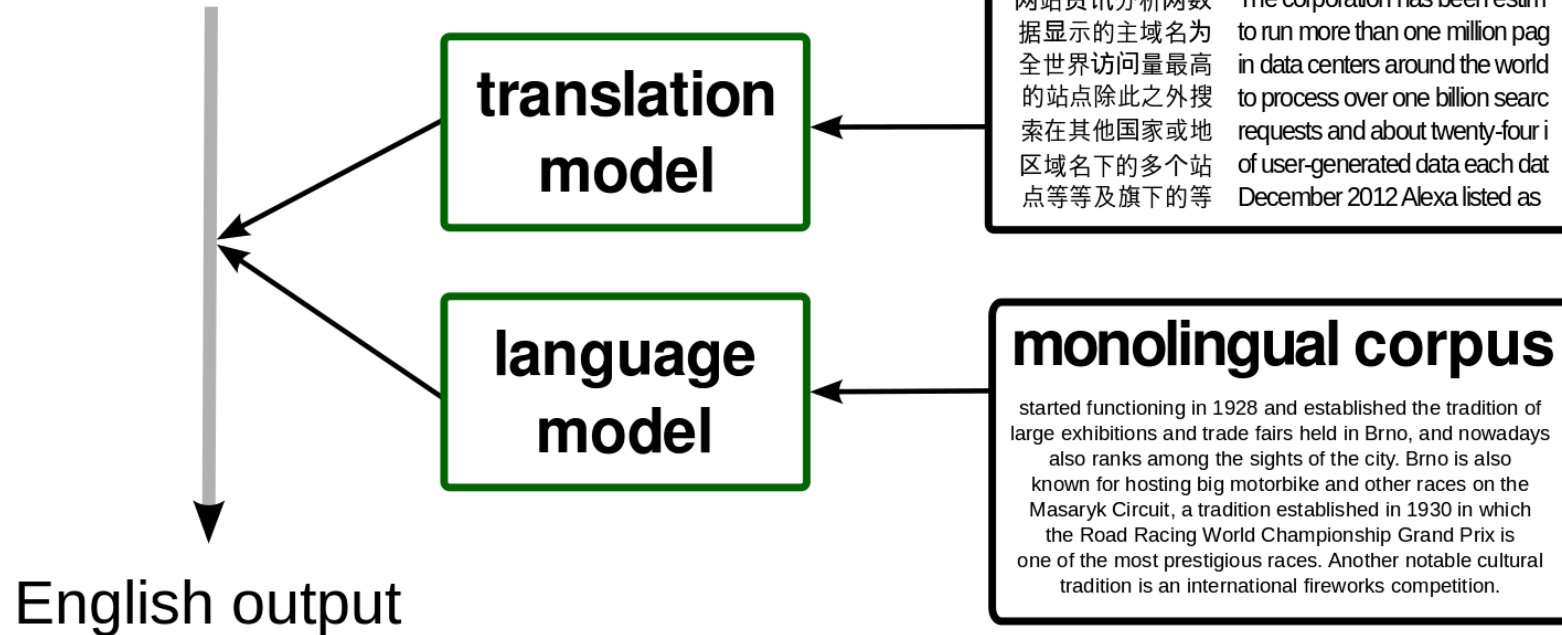
- Fred Jelinek



Statistical Machine Translation

- Required a parallel corpora

似乎格式有問題



Rise of statistical learning

- Required human-designed representations and input features

Var	Definition	Value in Fig. 5.2
x_1	count(positive lexicon) $\in$ doc)	3
x_2	count(negative lexicon) $\in$ doc)	2
x_3	$\begin{cases} 1 & \text{if "no"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	1
x_4	count(1st and 2nd pronouns $\in$ doc)	3
x_5	$\begin{cases} 1 & \text{if "!"} \in \text{doc} \\ 0 & \text{otherwise} \end{cases}$	0
x_6	log(word count of doc)	$\ln(64) = 4.15$

Can these features be automatically learned?



Promise of deep learning

- Most NLP works in the past focused on human-designed representations and input features

Var	Definition	Value in Fig. 5.2
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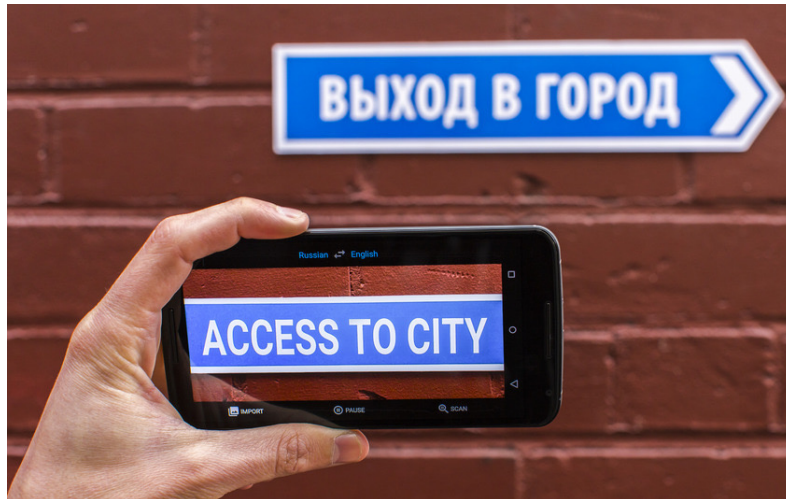
- **Representation learning attempts to automatically learn good features and representations**
- **Deep learning attempts to learn multiple levels of representation on increasing complexity/abstraction**



Deep learning era

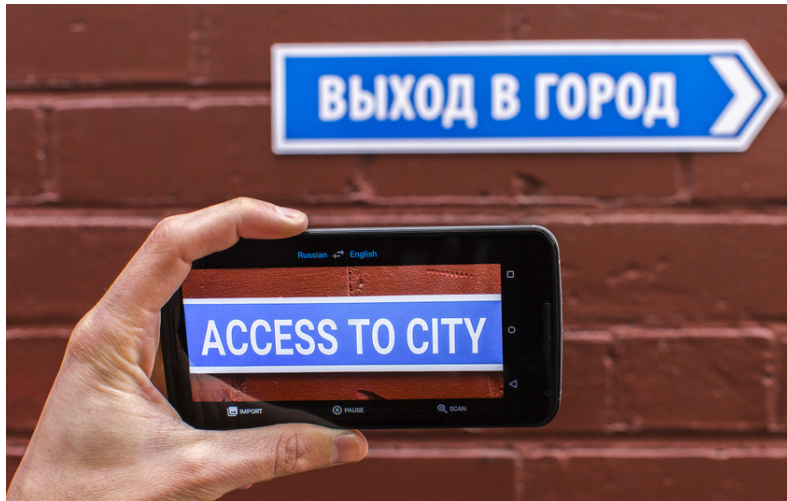
2014-Present: Neural Machine Translation

- Led to significant advances in core NLP technologies



Deep learning era

- Significant advances in core NLP technologies
- **Essential ingredient:** large-scale supervision, lots of compute
- Reduced manual effort - less/zero feature engineering



36M sentence pairs

Russian: Машинный перевод - это круто!



English: Machine translation is cool!



Machine Translation

BBC

Sign in

选项 (英文)

检索

Q

NEWS | 中文

繁

主页

国际

两岸

英国

评论

科技

财经

图辑

音频材料

视频材料

BBC英伦网

巴拿马首任驻华大使专访：与台湾断交之后

他说，不认同中国“买走”台湾邦交国的说法，与中国建立关系对巴拿马有利，不担心影响与美国关系。

🕒 1小时前

巴拿马外交转向周年 中美大国博弈内幕解密

尼加拉瓜运河成谜：人走楼空 “不再提及”

触发萨尔瓦多与台湾断交的港口令美国担忧





人民币贬值、楼价高涨：消费降级中产失去的优质生活

广告



台湾民间发起东京奥运“正名”公投的意义





观点：高铁“一地两检”——中国的强势 港人的无力

江浙：两地价值观和制度差异巨大，冲突无可避免。香港没有反对的权利，也没有反对的能力。



范冰冰消失百日后 中国娱乐业的寒蝉效应

日前发布的《中国影视明星社会责任研究报告》中，最高分徐峥为78分，最低分范冰冰为0分。

🕒 2小时前

中蒙参加俄罗斯军演“中俄靠拢论”再吸眼球

🕒 2018年 9月 11日

日本人脚踹慰安妇铜像引爆台湾人抗议

🕒 2018年 9月 11日

太空望远镜探测比“三

Machine Translation

BBC Sign in Option (English) 检索

NEWS | 中文 Traditional

Homepage International Cross-strait United Kingdom comment Technology Finance Picture series Audio material

Interview with Panama's first ambassador to China: After breaking diplomatic relations with Taiwan

He said that he does not agree with China's saying that "buy" Taiwan's diplomatic relations with China. Establishing relations with China is beneficial to Panama and does not worry about affecting relations with the United States.

🕒 1 hour ago

Panamanian Diplomacy Turns to Anniversary

The Nicaragua Canal is a mystery: people go to the floor and "no longer mention"

The port that triggered the break of El Salvador and Taiwan has worried the United States



Renminbi depreciation, property prices are rising: consumption downgrades the loss of quality life in the middle class

ADVERTISING



The significance of the Taiwanese people's "referred to" referendum



China and Mongolia participate in the Russian military



Google's Neural Machine Translation System: Bridging the Gap between Human and Machine Translation

<https://ai.googleblog.com/2016/09/a-neural-network-for-machine.html>

<i>Input sentence:</i>	<i>Translation (PBMT):</i>	<i>Translation (GNMT):</i>	<i>Translation (human):</i>
李克強此行將啟動中加總理年度對話機制，與加拿大總理杜魯多舉行兩國總理首次年度對話。	Li Keqiang premier added this line to start the annual dialogue mechanism with the Canadian Prime Minister Trudeau two prime ministers held its first annual session.	Li Keqiang will start the annual dialogue mechanism with Prime Minister Trudeau of Canada and hold the first annual dialogue between the two premiers.	Li Keqiang will initiate the annual dialogue mechanism between premiers of China and Canada during this visit, and hold the first annual dialogue with Premier Trudeau of Canada.

(Wu et al., 2016)



Google's Neural Machine Translation System: Bridging the Gap between Human and Machine Translation

<https://ai.googleblog.com/2016/09/a-neural-network-for-machine.html>

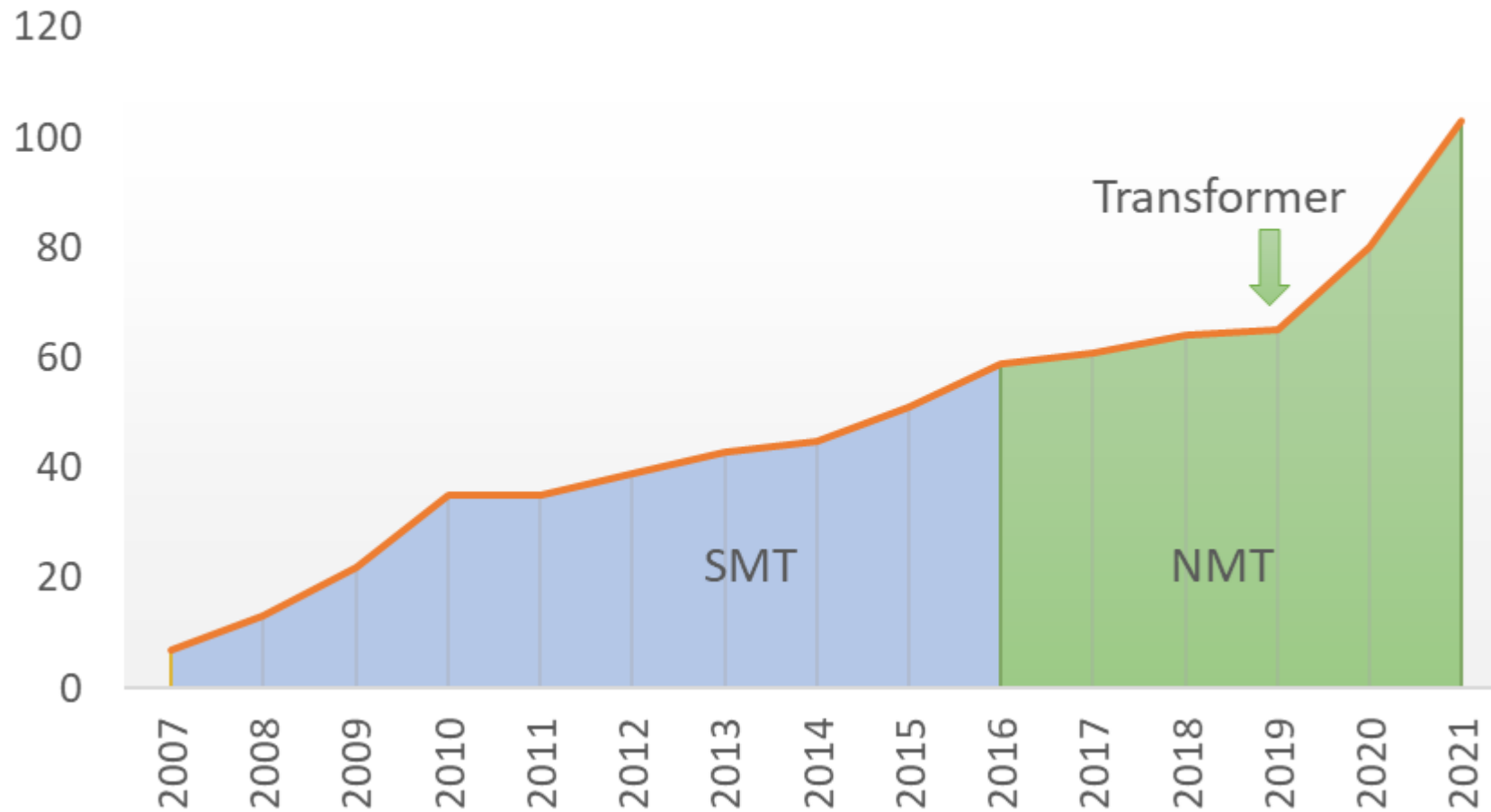
Table 10: Mean of side-by-side scores on production data

	PBMT	GNMT	Human	Relative Improvement
English → Spanish	4.885	5.428	5.504	87%
English → French	4.932	5.295	5.496	64%
English → Chinese	4.035	4.594	4.987	58%
Spanish → English	4.872	5.187	5.372	63%
French → English	5.046	5.343	5.404	83%
Chinese → English	3.694	4.263	4.636	60%

(Wu et al., 2016)



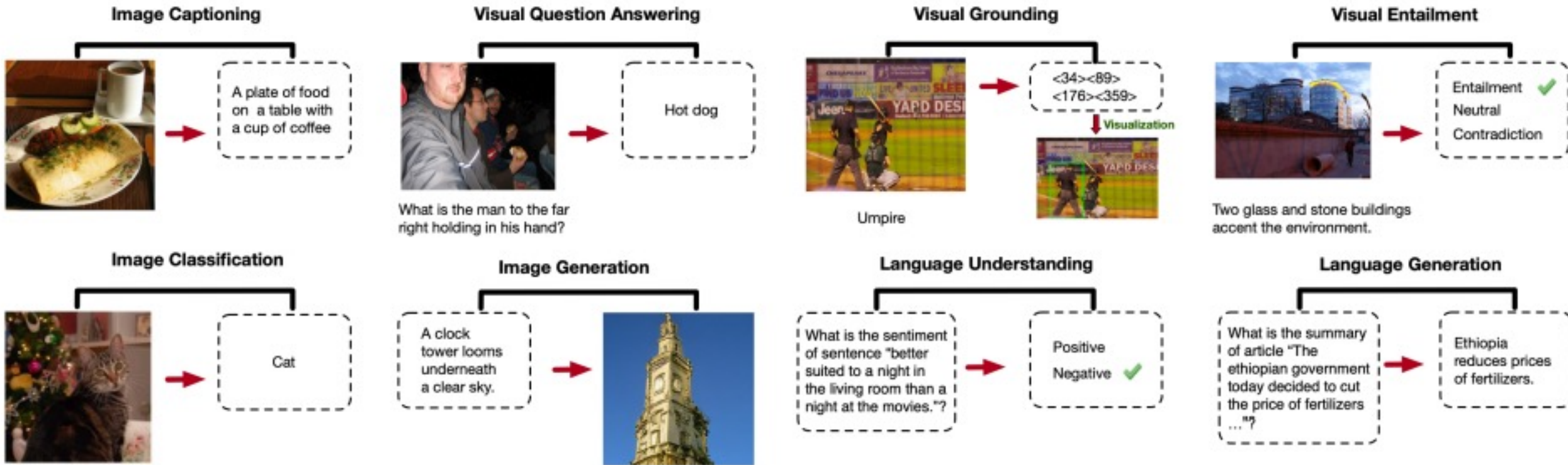
Machine Translation (across 100 languages)



<https://www.microsoft.com/en-us/research/blog/microsoft-translator-now-translating-100-languages-and-counting>



Multimodal models



<https://github.com/OFA-Sys/OFA>

OFA (Wang et al, 2022)



Power of Data

CleverBot (2010)

User:	Do you like Doctor Who?
Cleverbot:	Yes, so funny.
User:	What's your favorite show?
Cleverbot:	I don't have one.
User:	What's your name?
Cleverbot:	Jason I already told you.
User:	No your real name.
Cleverbot:	Stephanie IS my real name.
User:	I thought it was Jason?
Cleverbot:	How did you come to that conclusion.
User:	Because you said so.
Cleverbot:	I have to do my homework.

<https://www.cleverbot.com/>

How it works:

- Corpus of conversational turns
- Find the most similar sentence and copy the response
- Learn from human input

What do you get?

- Something that someone say
- Incoherent conversation



Power of Data

Meena (Google, 2020)

Conversation D

Human: *I like how cute cows are*
Meena: *Ah, interesting. You like how they are cute, right?*
Human: *yeah, and they're really smart*
Meena: *I heard they go to college*
Human: *cows go to college?*
Meena: *I heard that a cow went to Harvard.*
Human: *what did the cow study?*
Meena: *Bovine sciences.*
Human: *do horses go to Harvard?*
Meena: *Horses go to Hayvard.*
Human: *that's a pretty good joke, I feel like you led me into it*
Meena: *You were trying to steer it elsewhere, I can see it.*

How it works:

- Corpus of conversational turns (over 40B words)
- Train huge neural network (2.6 billion parameters) for 30 days on 2048 TPUs cores
- Predict response given a sentence



Power of Data

LaMDA (Google, 2022)

Hi! Hello, I am a friendly dialog model. What do you want to talk about?

What's a good topic for a new blog?

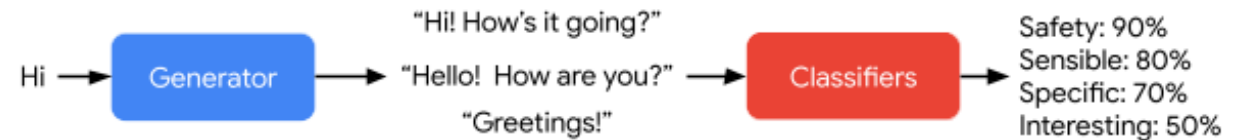
Well there are so many! How about something about a new food item that you just tried.

There are lots of food blogs. How can I make it unique?

Hmmm, you can write about a food that you tried somewhere that is exclusive to that location or restaurant.

How it works:

- Pretrain on even more data
- Fine-tune so it has certain characteristics



Is this chatbot sentient?

<https://ai.googleblog.com/2022/01/lamda-towards-safe-grounded-and-high.html>

https://youtu.be/CvD9U_3hBPo



Power of Data

ChatGPT (OpenAI, 2022)

<https://openai.com/blog/chatgpt/>



write a sonnet about pokemon



A world of wonder, filled with creatures wild,
A realm of magic, where adventures start,
From Kanto to Kalos, far and wide,
The world of Pokémon, alive and heart.

With Pikachu as guide, we set out to roam,
To catch and train them all, to be the best,
Battling Gym Leaders, overcoming foes,
Our bond with creatures strong, a bond we'll never test.

But it's not just battles that define,
The world of Pokémon, so full of cheer,
It's friendships formed, and lessons learned,
A world that's full of love, and never fear.

So let's go on a journey, you and I,
Through the world of Pokémon, beneath the sky.

How it works:

- Trained on conversational data
- Uses reinforcement learning with human feedback

The Brilliance and Weirdness of ChatGPT

A new chatbot from OpenAI is inspiring awe, fear, stunts and attempts to circumvent its guardrails.

<https://www.nytimes.com/2022/12/05/technology/chatgpt-ai-twitter.html>

Time taken to reach 1m users (mths)

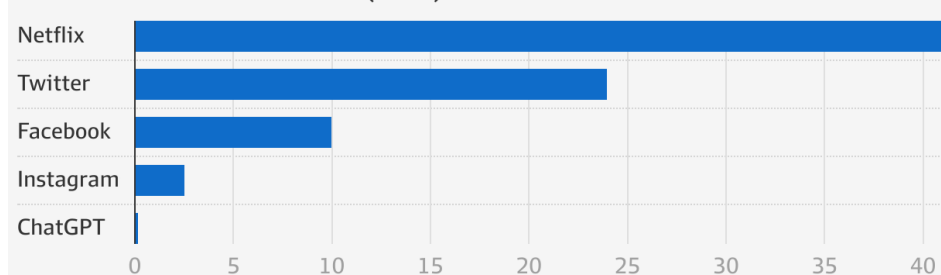


Chart: Financial Review • Source: Genevieve Roch-Decter, CFA

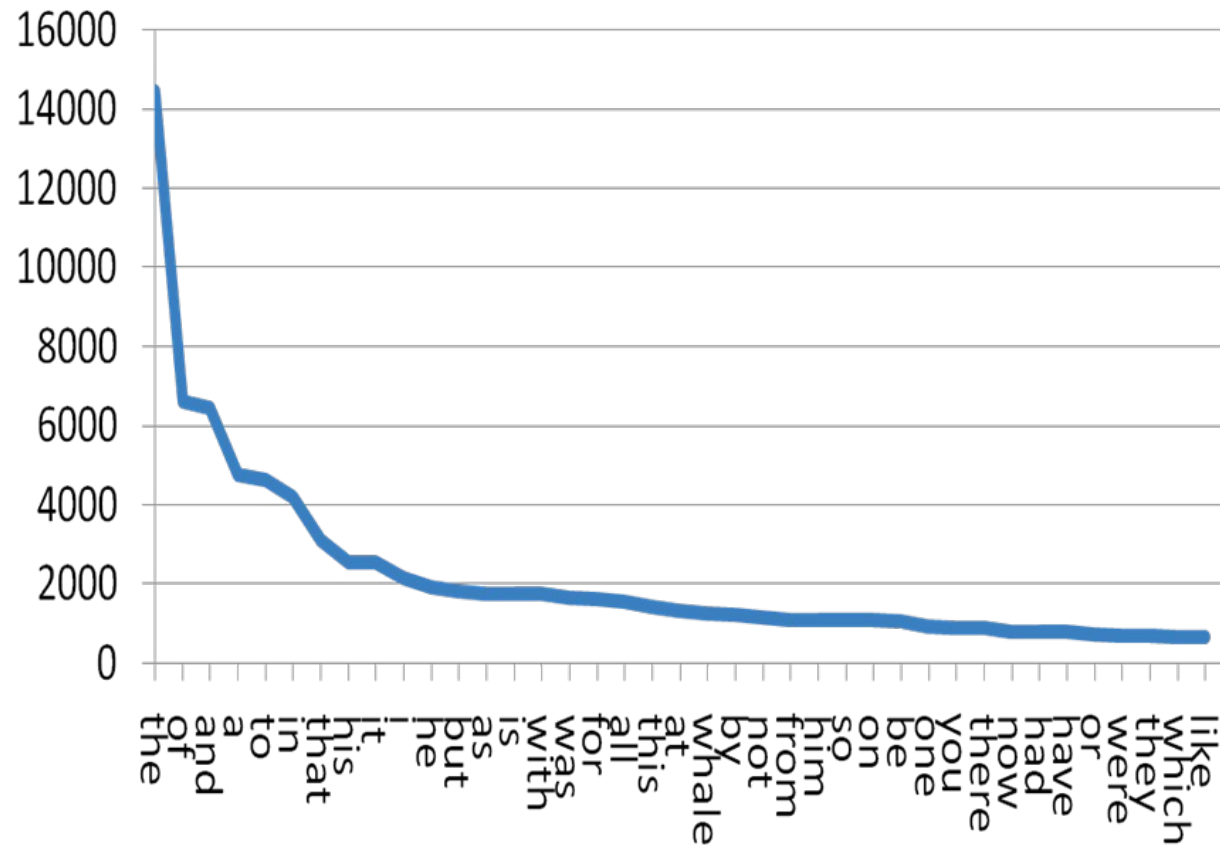


Many remaining challenges in NLP!



Challenges in modern NLP

- *Scale*: Large number of phenomena
- *Sparsity*: Text data is often heavy-tailed



Ambiguity in grounding

I saw her duck



Challenges in modern NLP

- *Bias*: Models learn biases in available data

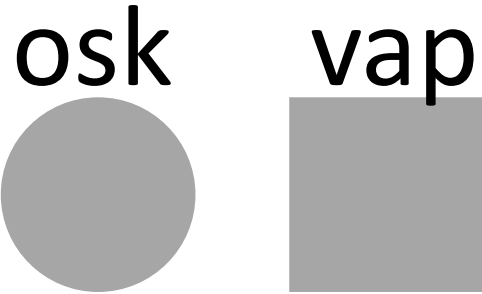


- *Context:* Knowledge bases, perception, interaction



Compositional generalization

Grounding



Compositionality



Generalization

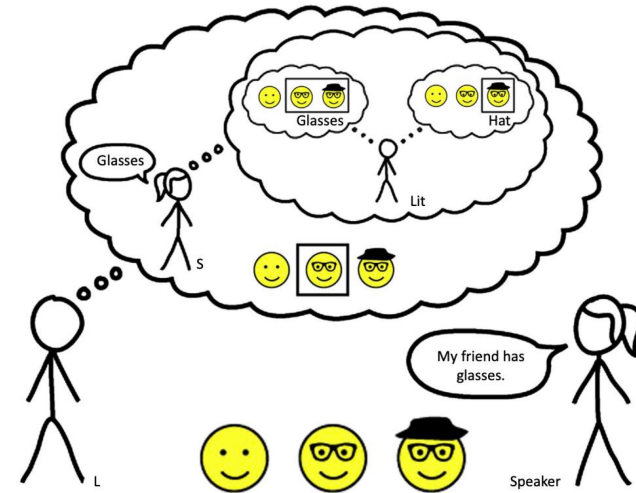


Challenges of understanding language

- Does solving language require solving AI?
- Language derives its meaning from use



'block' 'pillar' 'slab' 'beam'.



- To truly understand language: need **context**, **world knowledge**, **mental model** of other people





ALFRED

A Benchmark for Interpreting
Grounded Instructions for Everyday Tasks

<https://askforalfred.com/>



What does it mean to understand language?
A look at levels of linguistic representations



Translation

Dream of universal translator

- Different languages: Machine translation

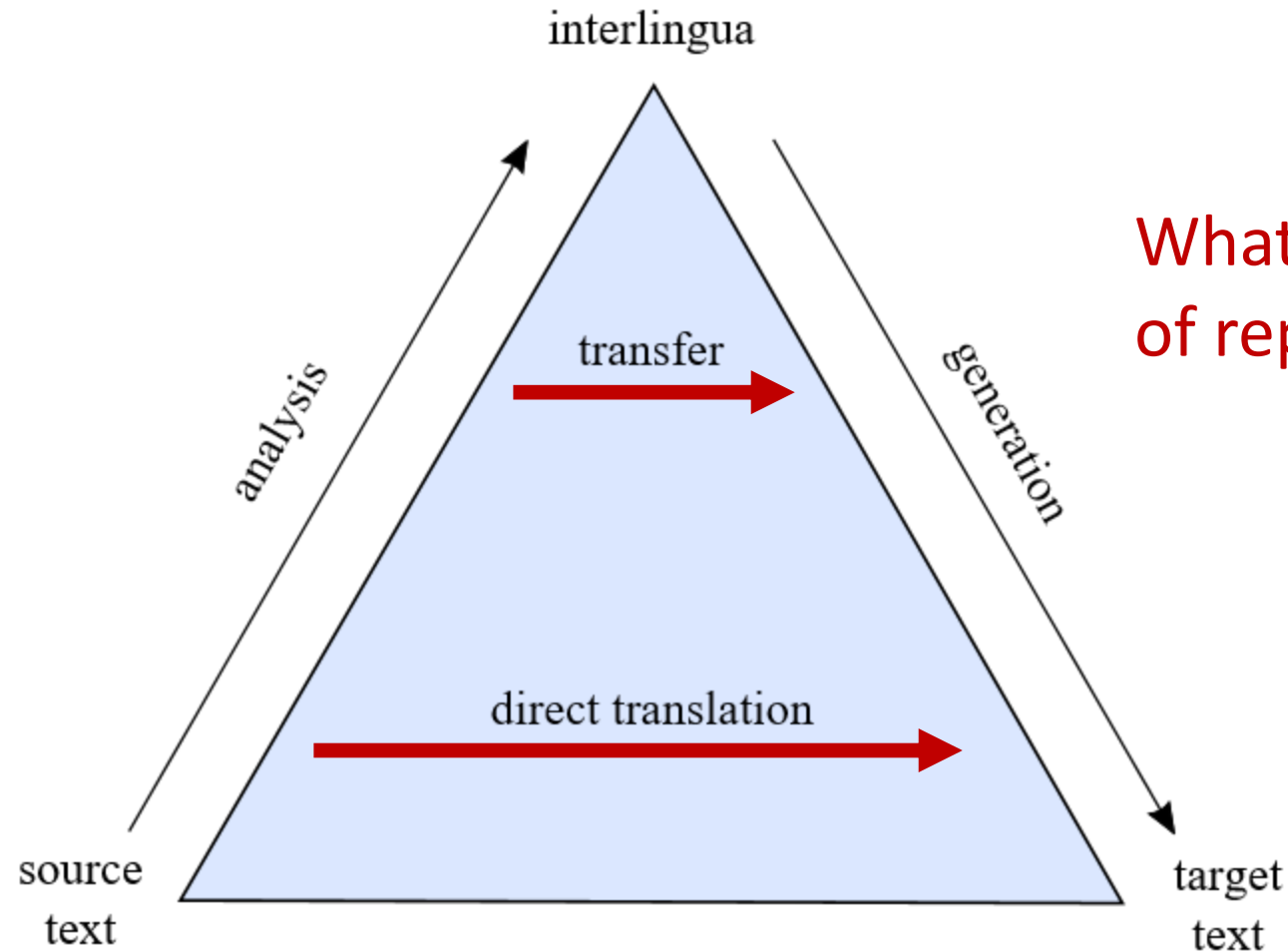
the cat sat on the table  η γάτα κάθισε στο τραπέζι

Not just between languages, but across modalities as well

- Audio to text: Speech recognition
- Image to language: Image captioning



Vauquois Triangle for translation



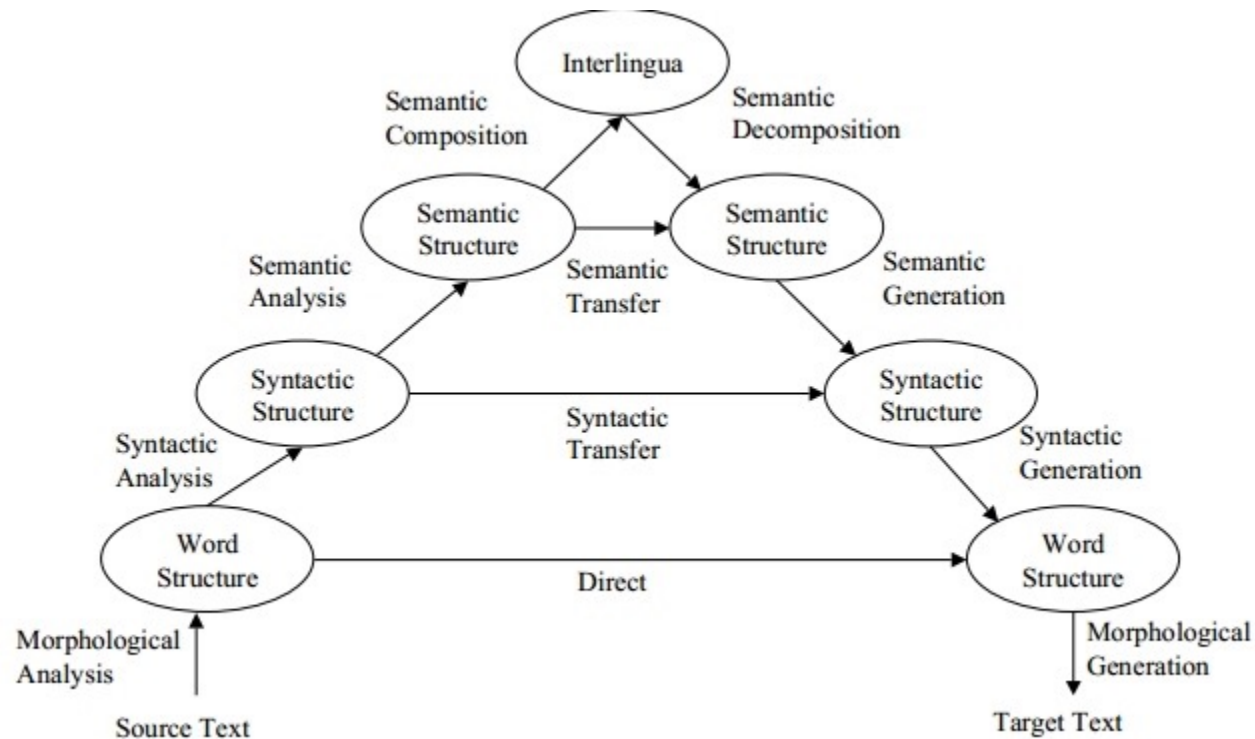
What are the levels of representation?

the cat sat on the table

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Vauquois Triangle for translation



Pragmatics

Semantics

Syntax

Morphology

the cat sat on the table

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Levels of language

- Phonetics: acoustic and perceptual elements
- Phonology: basic sounds (phonemes) and rules for combination
 - e.g. syllable constraints in Japanese.
 - e.g. English loanword for “dog” in Japanese is “doggu”
- Morphology: how morphemes combine to form words, relationship of phonemes to meaning
 - e.g. delight-ed vs. de-light-ed

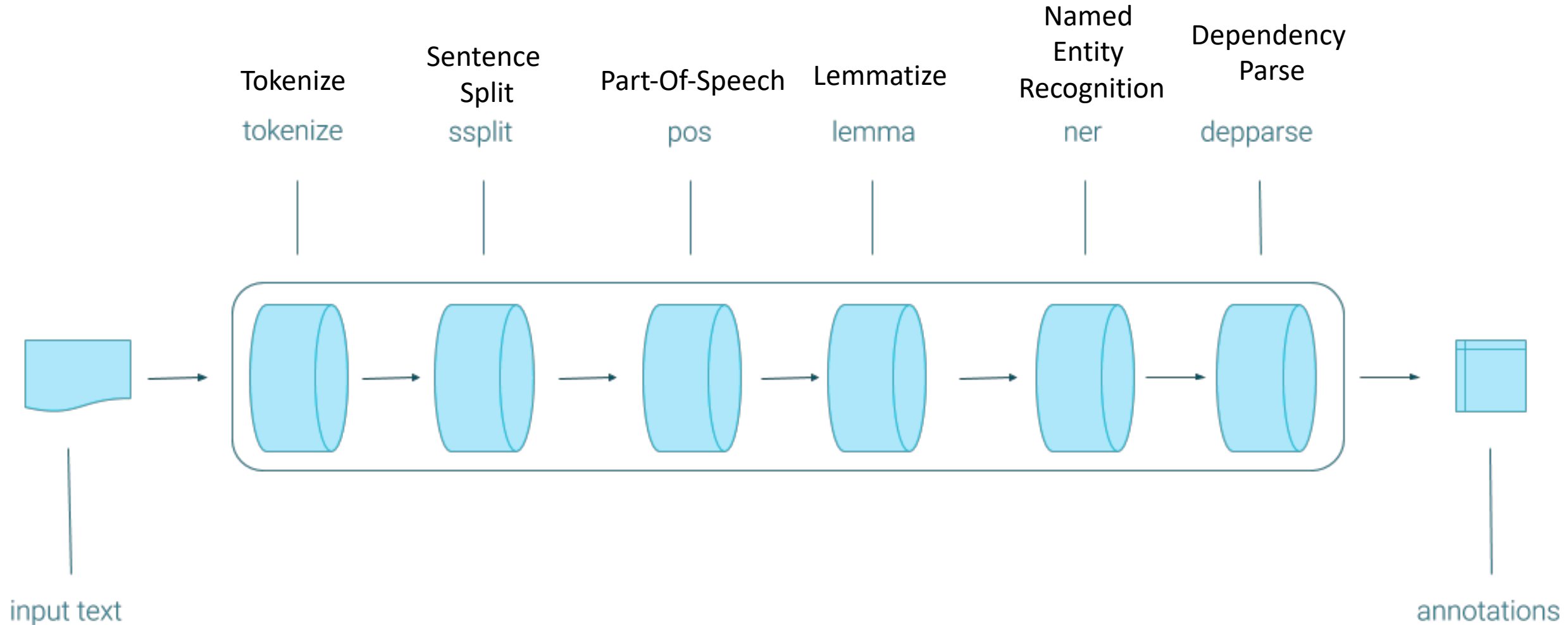


Levels of language

- Syntax: set of rules, principles, and processes that govern the structure of sentences in a given language
 - sentence formation
 - o e.g. The clown who the musician hits watches the ballerina
- Semantics: focuses on the relation between *signifiers*, like words, phrases, signs, and symbols, and what they stand for; their denotation
 - meaning (from syntax to logical formulas)
 - e.g. Everyone is not here => what does this mean? Nobody / Not everyone is here.
- Pragmatics: studies the ways in which context contributes to meaning
 - e.g. “The ham sandwich wants another coffee”



Traditional NLP Pipeline



More details in optional video lecture (to be posted on Coursys)

Annie, the kitten was born at night. She was a gray furball.



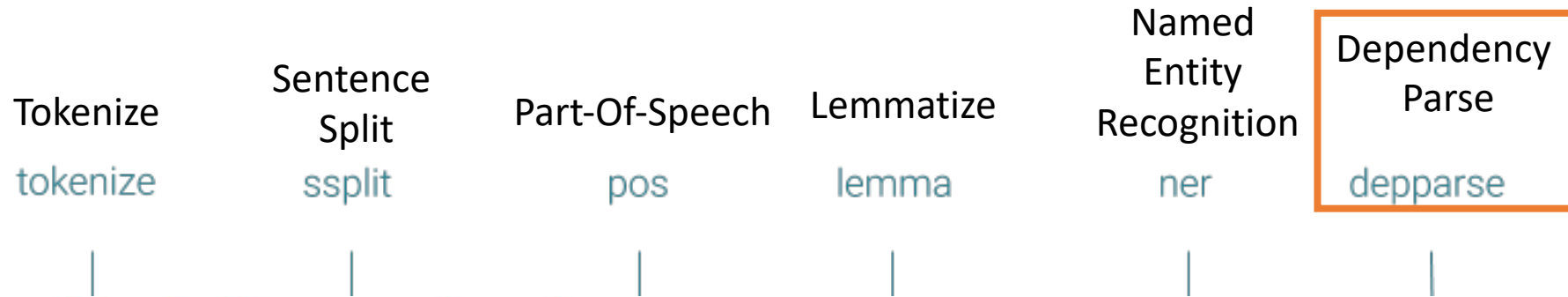
Traditional NLP Pipeline



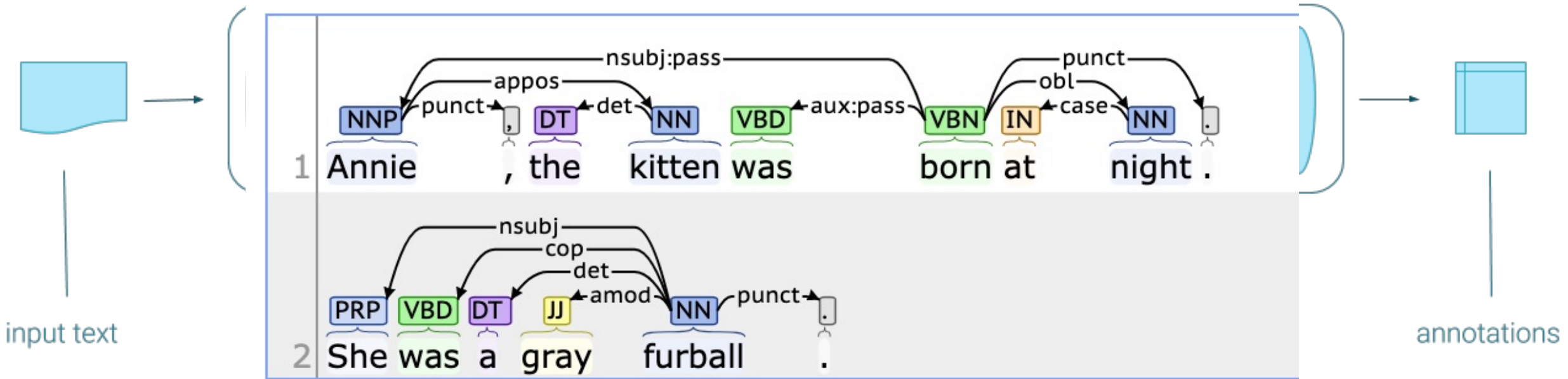
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Traditional NLP Pipeline



Basic Dependencies:



More details in optional video lecture (to be posted on Coursys)

Annie, the kitten was born at night. She was a gray furball.

Try out some NLP systems for yourself!

- Spacy (<https://spacy.io/>)
 - Interoperates with Stanza and AllenNLP
- Stanford CoreNLP / Stanza
 - CoreNLP (<http://stanfordnlp.github.io/CoreNLP/>)
 - Stanza (<https://stanfordnlp.github.io/stanza>)
- AllenNLP (<https://allennlp.org/>)

What will this course cover?



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Representations

- Embeddings
- Compositional
- Structured

Methods

- Rule based
- Statistical Methods
- Neural Models
- Dynamic programming
- Language Models

Applications and Tasks

- Machine Translation
- Dialogue Agents
- Information Extraction
- Question Answering
- Grounding



What will this course cover?

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Methods

- Rule based
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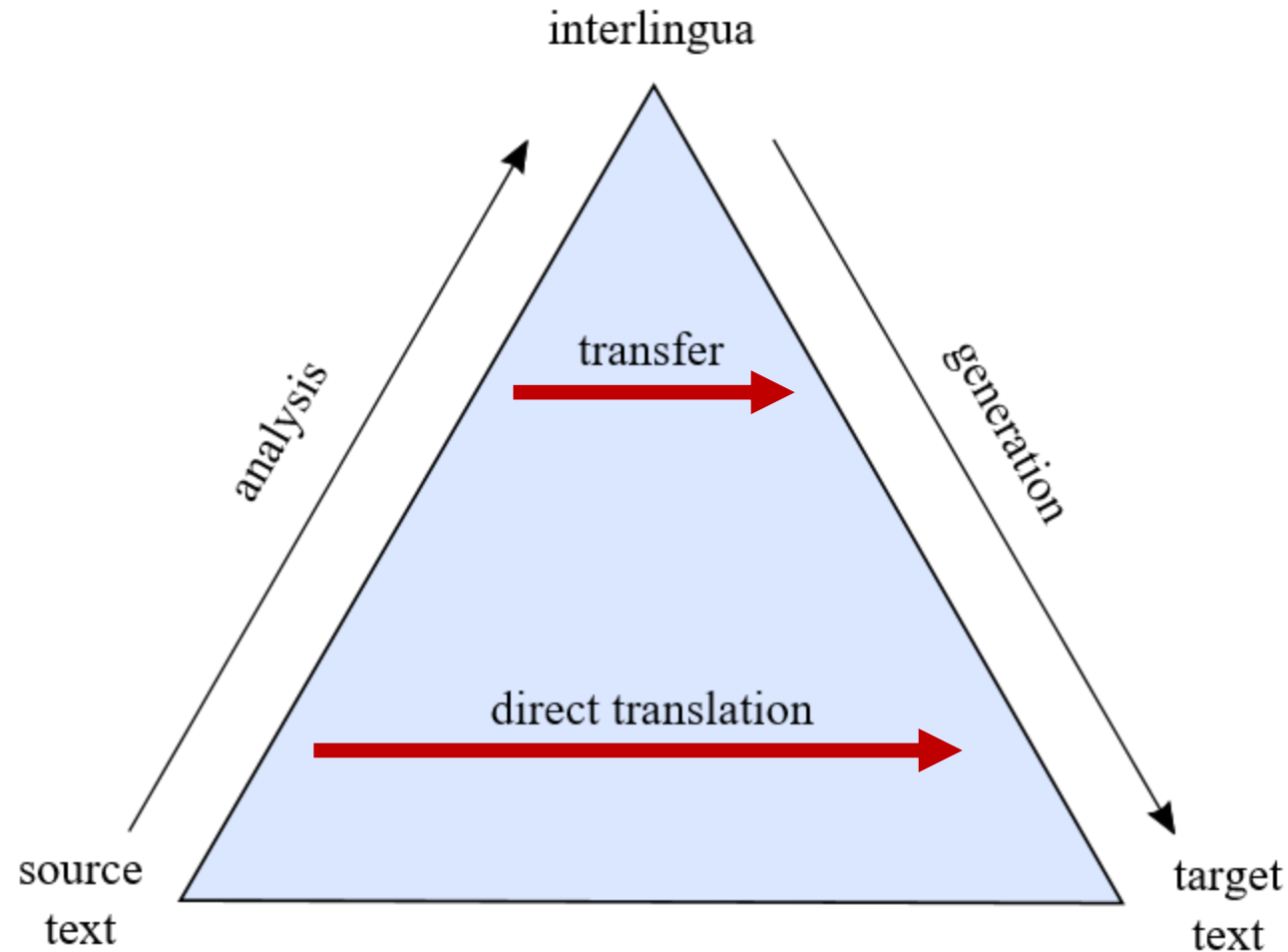
Applications and Tasks

- Machine Translation
- Dialogue Agents
- Information Extraction
- Question Answering
- Grounding

later part of the course



Vauquois Triangle for translation



the cat sat on the table

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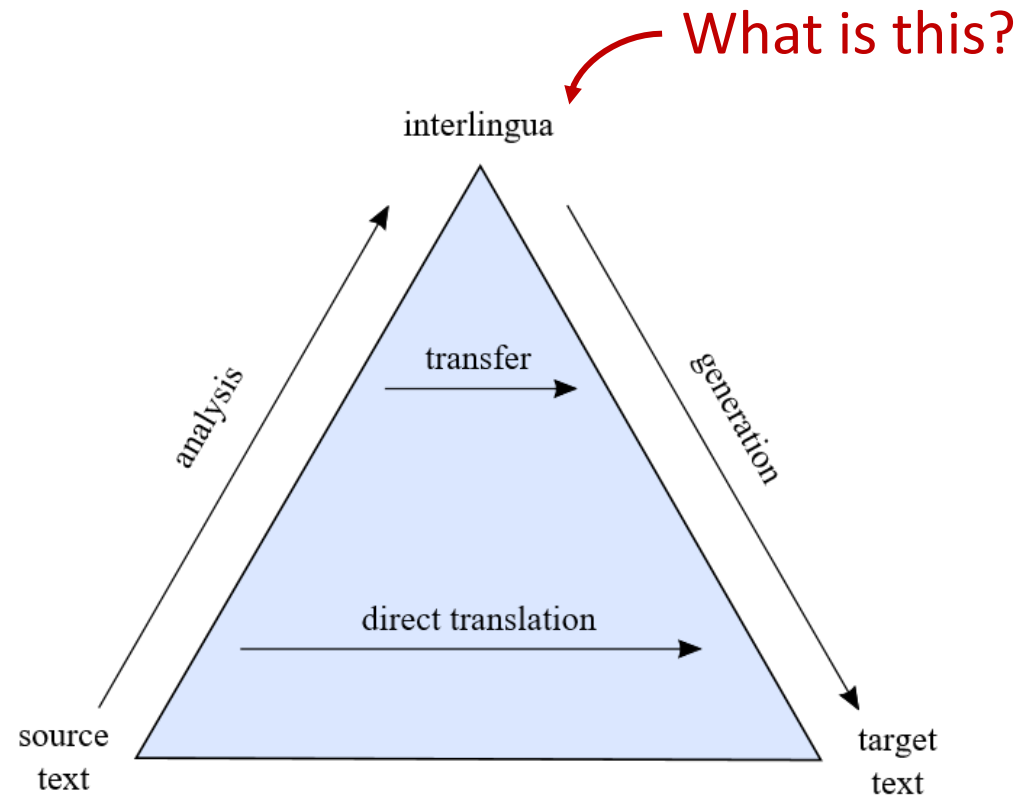


Key Questions

- How to **represent** language?
- How to **compose** meaning?
- How do we translate from the “surface form” of language to a representation that a computer can process, and back?
 - How do we **encode** / **parse** the meaning of a sentence?
 - How do we **decode** / **generate** text from a meaning representation?
- What are the key methods / algorithms for **learning** these representations?

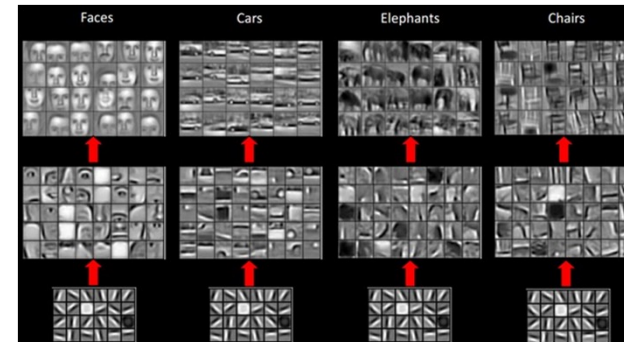


What should the representation look like?

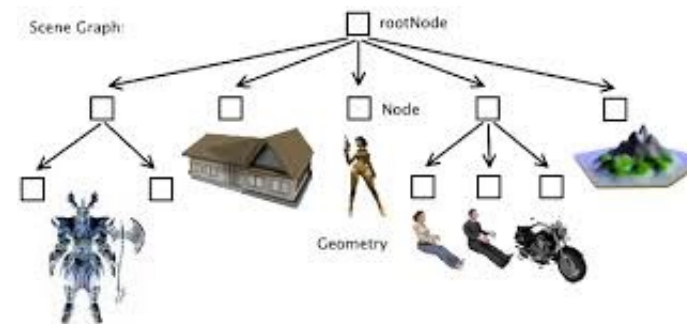


$$L = \begin{bmatrix} \vdots & \vdots & \vdots & \dots & \vdots & \vdots \\ \vdots & \vdots & \vdots & \dots & \vdots & \vdots \end{bmatrix}_n$$

the cat mat ...



Lambda expression:
 $((\lambda x . (\lambda y . x)) (\lambda z . z)) (\lambda a . a)$



Representations

How to represent the meaning of something?



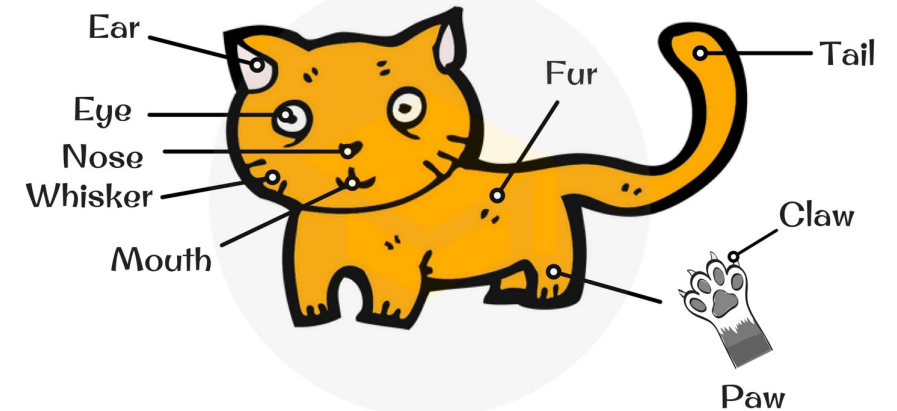
“cat”

cat: a small domesticated carnivore, *Felis domestica* or *F. catus*, bred in a number of varieties.

```
cat → {  
  isMammal: true  
  hasFur: true  
  hasLegs: true  
  meows: true  
  barks: false  
  height: 9.1 – 9.8 in  
  weight: 7.9 – 9.9 lbs  
  ...  
}
```

Attributed
representation

Parts of a cat



TESL.COM



Representations



“cat”



“dog”

Representing meaning as vectors

- common representation space
- enables information sharing
- can be learned from data

- One-hot

cat = [0 0 0 1 0 0 0]

dog = [0 0 0 0 0 1 0]

- Embeddings

cat = [0.04 1.79 -1.79 1.07 0.48]

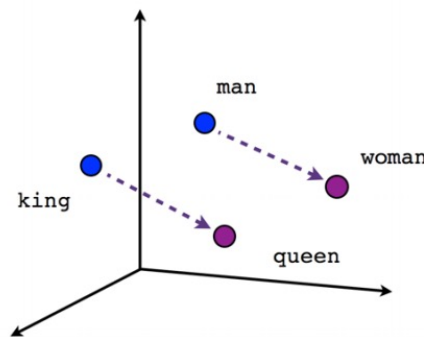
dog = [0.61 1.84 -1.12 0.52 0.53]



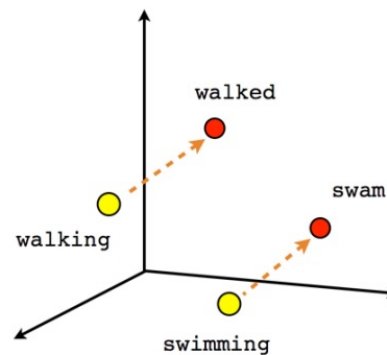
Distributed representations

Project words onto a continuous vector space

Similar words closer to each other



Male-Female

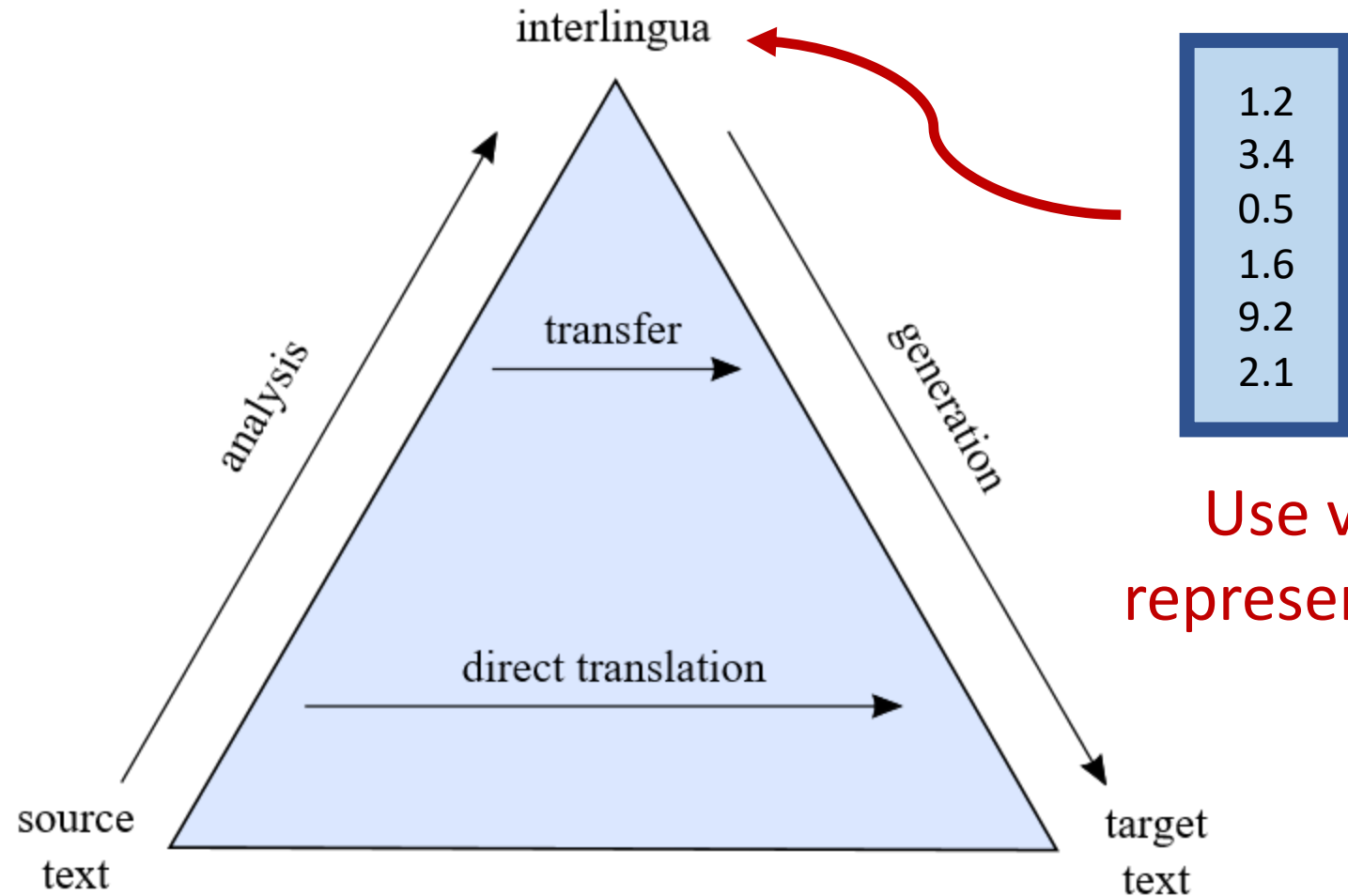


Verb tense



$$v(\text{king}) - v(\text{man}) + v(\text{woman}) = v(\text{queen})$$

Vauquois Triangle for translation



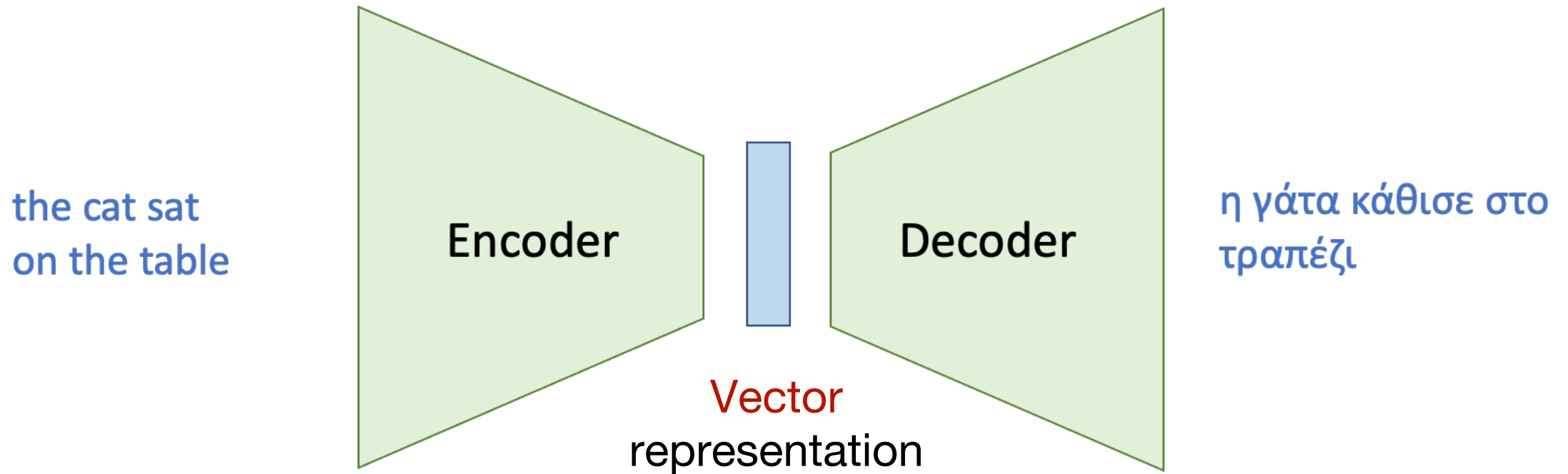
Use vector to
represent meaning

the cat sat on the table

η γάτα κάθισε στο τραπέζι



Vector based representation



Common choice: flat unstructured vector of real numbers

- easy to work with
- well defined mathematical operations
- can be used to measure if two things are similar or not



Translating across modalities

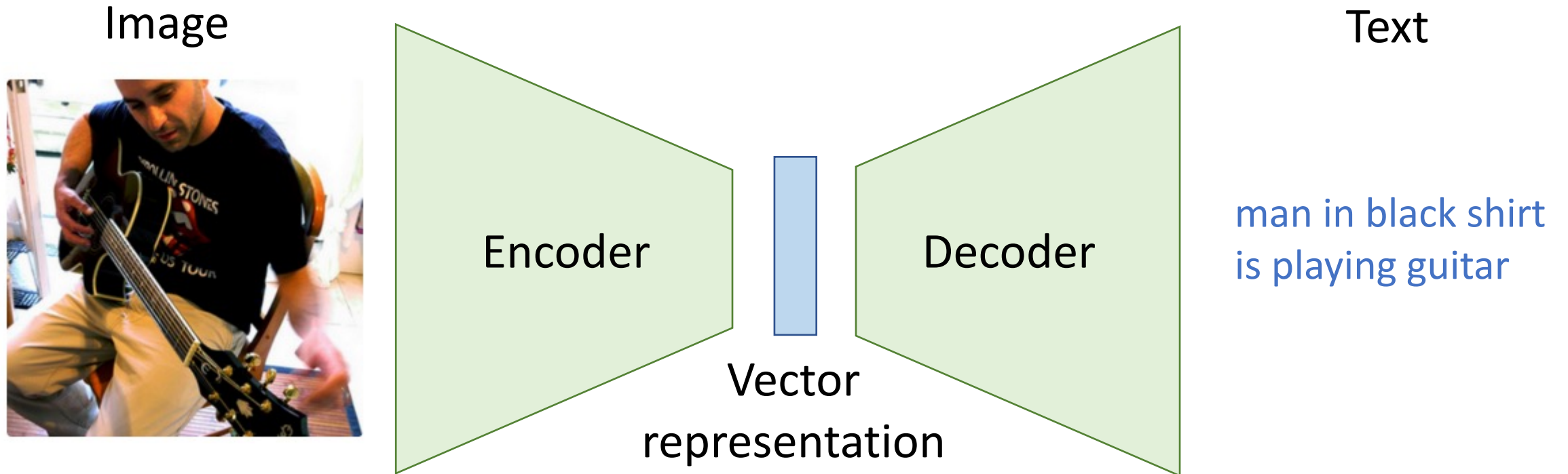
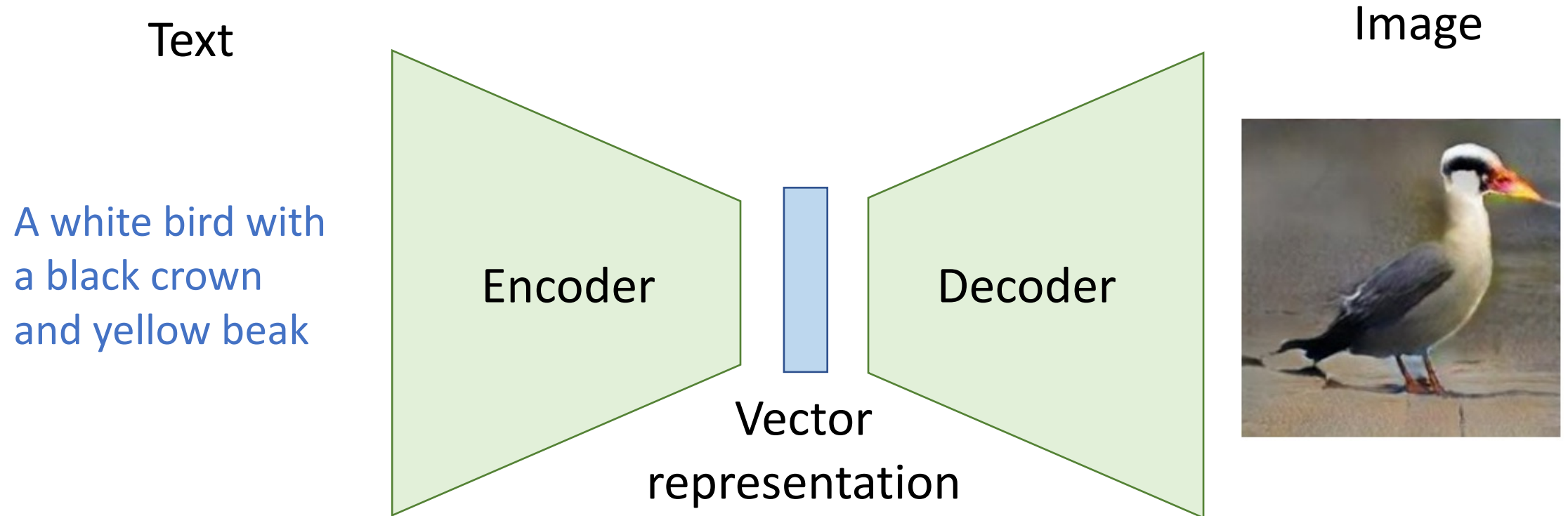


Image captioning

“Deep Visual-Semantic Alignments for Generating Image Descriptions”
[Karpathy and Fei-Fei CVPR 2015]



Translating across modalities



“StackGAN: Text to Photo-realistic Image Synthesis with Stacked Generative Adversarial Networks”
[Zhang et al, ICCV 2017]



Translating across modalities

Text

Brown colored dining table.
It has four legs made of wood.

Encoder

Vector
representation

Decoder

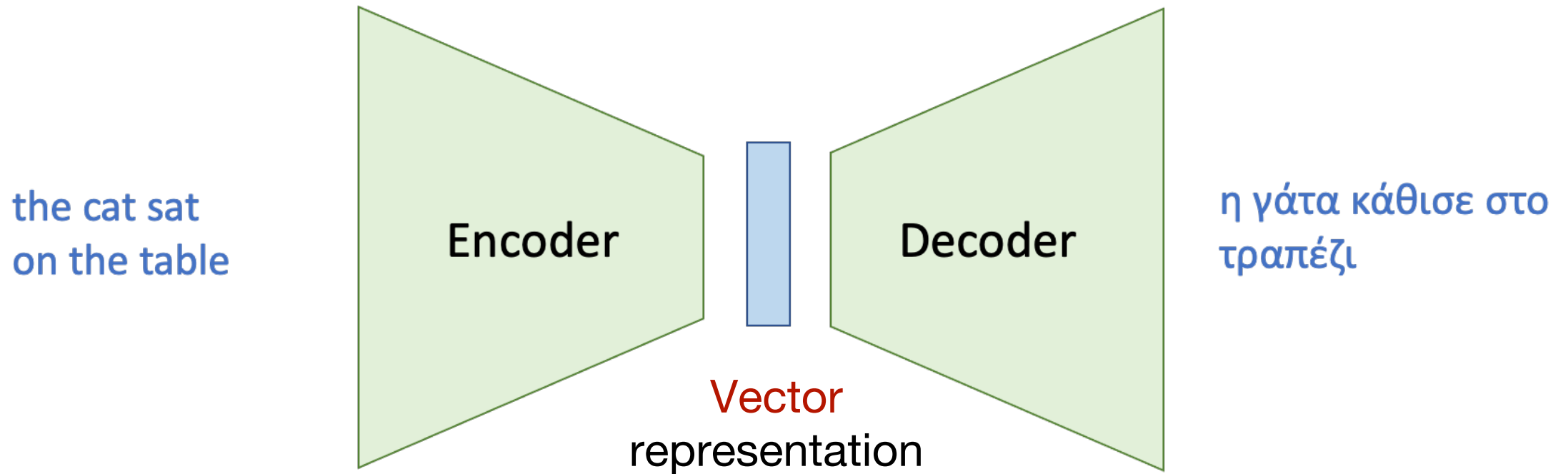
3D Shape



“Text2Shape: Generating Shapes from Natural Language by Learning Joint Embeddings”
[Chen et al, ACCV 2018]



Vector based representation



Common choice: flat unstructured vector of real numbers

- easy to work with
- well defined mathematical operations
- can be used to measure if two things are similar or not



How to **encode** language?

What is this function that takes the sentence and converts it into a vector?

the cat sat
on the table

Encoder

Vector
representation

Encoder

- How to convert the sentence into a meaning representation?

Understanding what is said
(encoding, parsing, feature extraction)

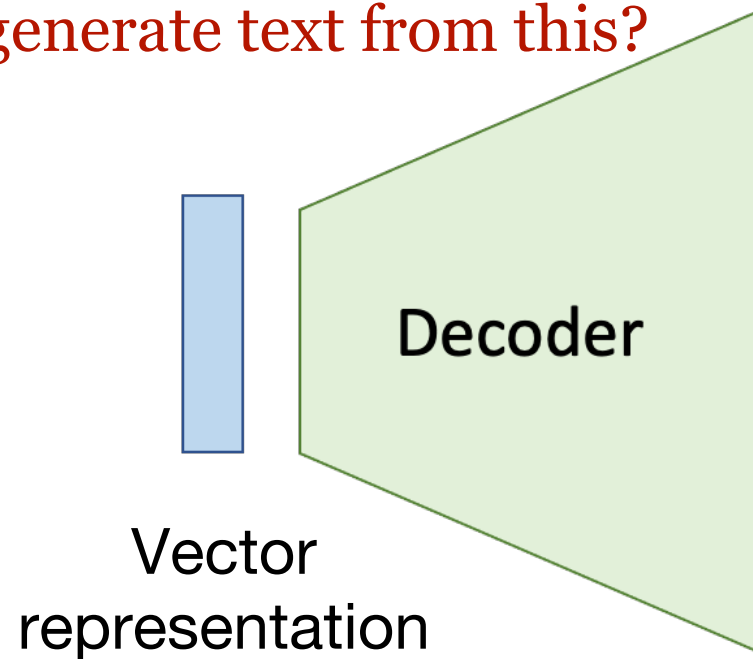


How to **decode** language?

How do we generate text from this?

Decoder

- How to take the vector representation and generate a sequence of words?



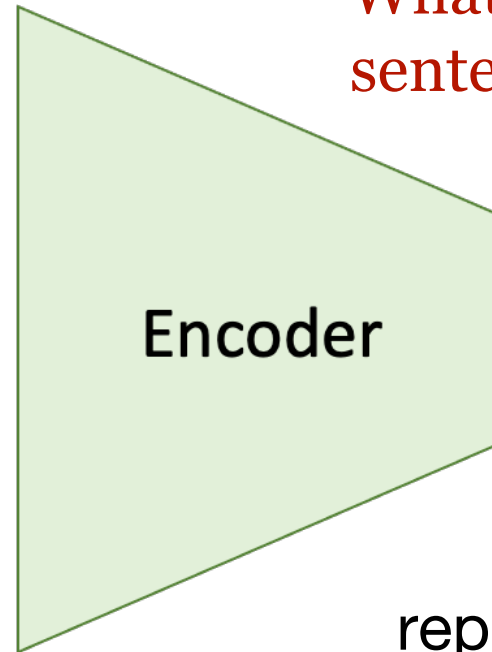
Deciding what to say
(decoding, generating)



How to **encode** language?

What is this function that takes the sentence and converts it into a vector?

the cat sat
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Vector
representation

Properties we want

- Be able to represent a single word
- Be able to represent the meaning of a sentence as a composition of the meaning of each word

Understanding what is said
(encoding, parsing, feature extraction)



How to learn representations for **words**?

use **context** to represent words



“cat”



“dog”

start with representing words as discrete tokens

- One-hot

cat = [0 0 0 1 0 0 0]

dog = [0 0 0 0 0 1 0]

learn about neural models that use vectors

- Embeddings

cat = [0.04 1.79 -1.79 1.07 0.48]

dog = [0.61 1.84 -1.12 0.52 0.53]

...government debt problems turning into **banking** crises as happened in 2009...
...saying that Europe needs unified **banking** regulation to replace the hodgepodge...
...India has just given its **banking** system a shot in the arm...

These **context words** will represent **banking**

learn a dense vector representation that can be used to **predict** words from their context

- Predict what goes next

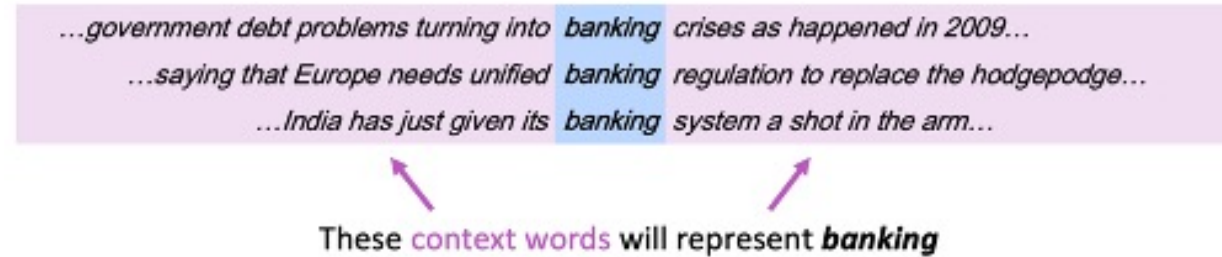
The cat sat on the _____

- Fill in the blank

A bottle of _____ is on the table.



How to learn representations



- Pretrain on large amount of text
- During pretraining: learn to predict words from their context

- Predict what goes next

The cat sat on the _____

Language Model $P(w_t | w_{<t})$

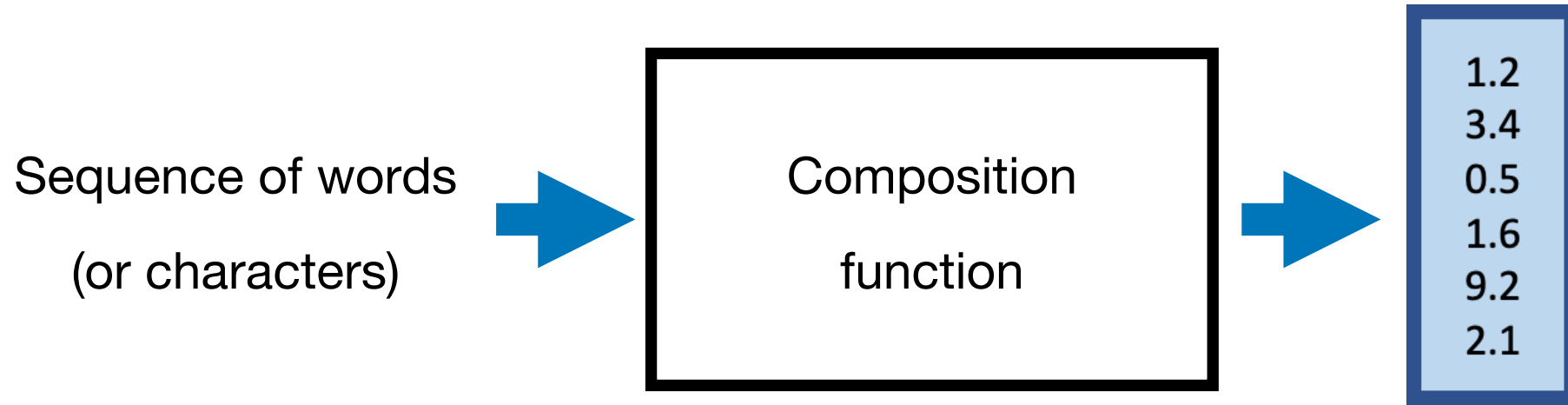
- Fill in the blank

A bottle of _____ is on the table.

- Use pretrained model and fine-tune it for downstream tasks



How to **compose** language?



Bag of Words (BOW)

Window BoW (bag of ngrams)

RNN

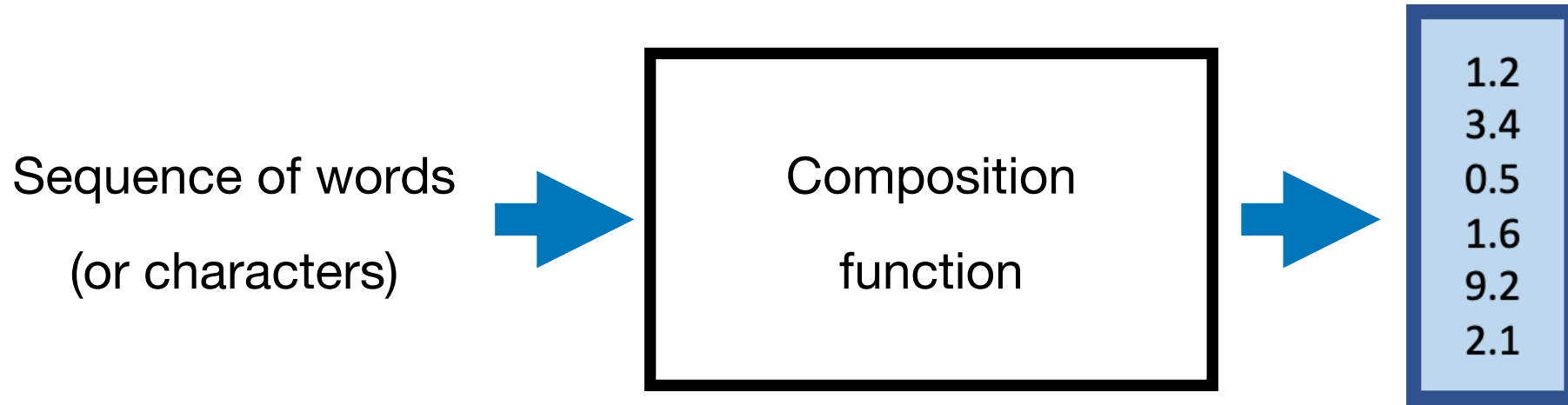
Transformers

CNN

GraphNN

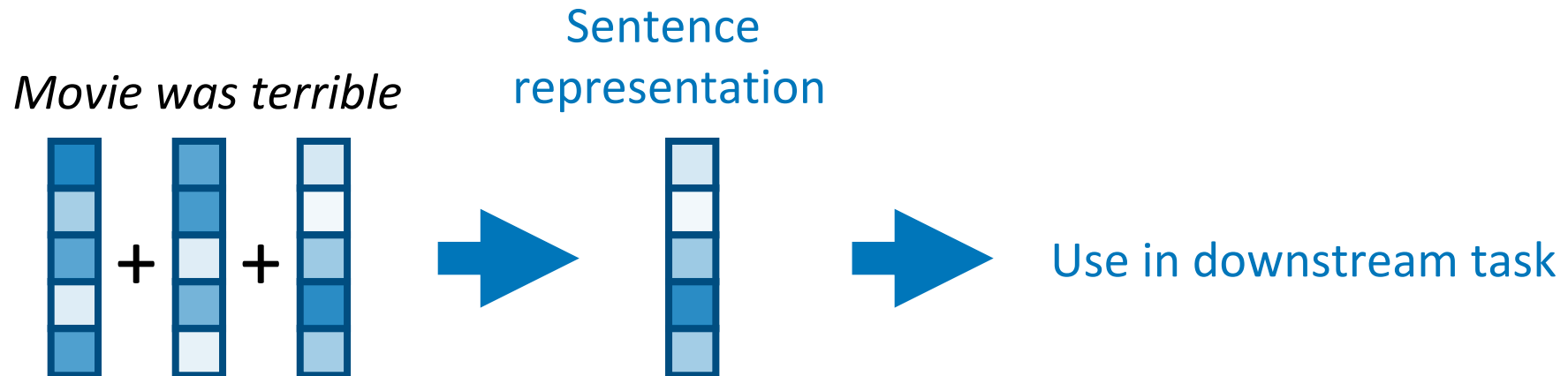


How to **compose** language?



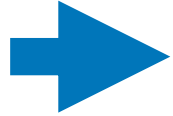
Simplest: take weighted average

Example:



How to use the learned **representation**?

What do we do with this vector representation after we get it?



Use in downstream task

Vector
representation

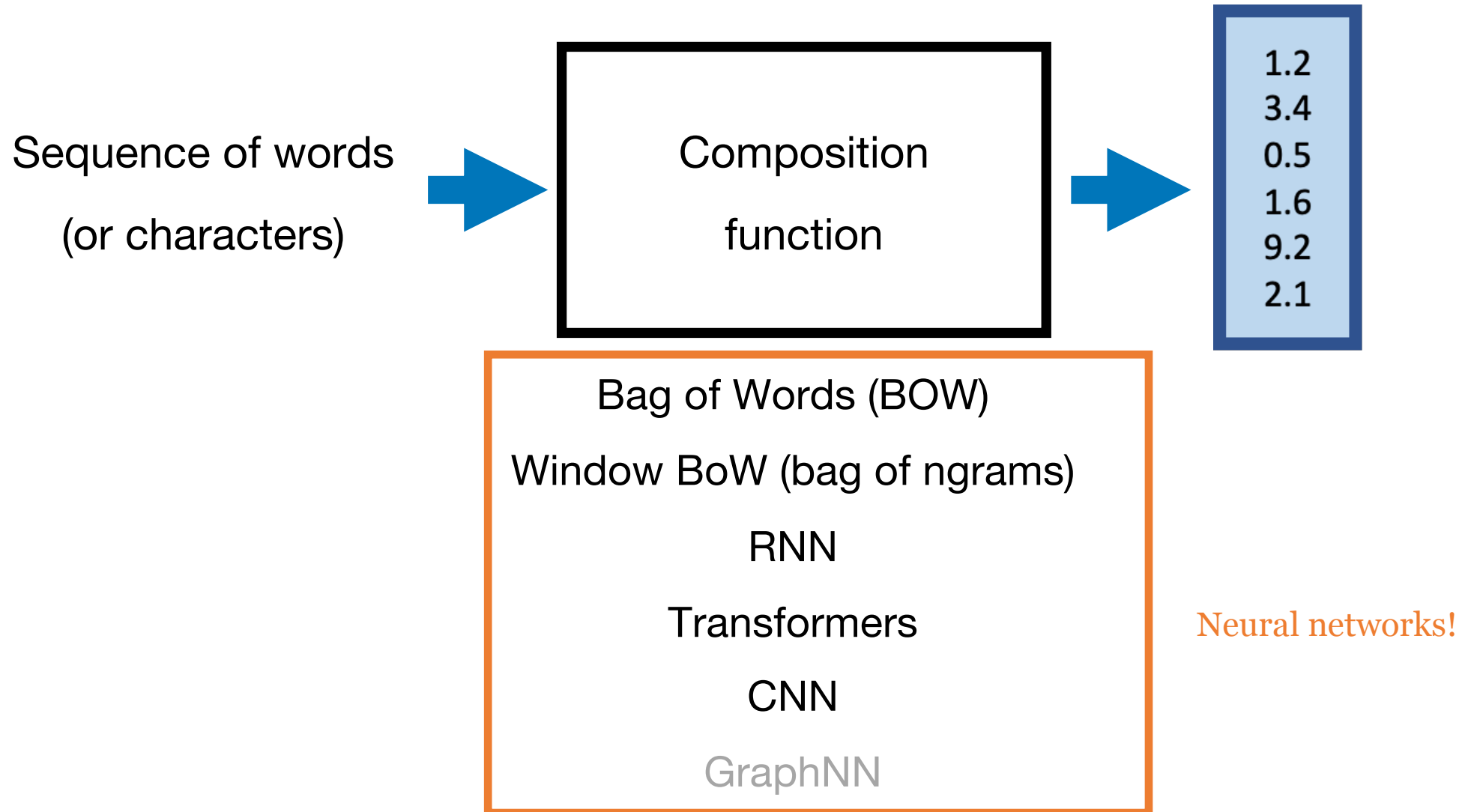
Do prediction: classify as spam/not spam

Use to generate text (also known as **decoding**)

Can also be used to generate other stuff (like images!)



How to **compose** language?



Thank you!



Getty Images

<https://www.thoughtco.com/syntactic-ambiguity-grammar-1692179>

